

4.8 LINKAGES OF MORE THAN FOUR BARS

With some exceptions,* the same approach as shown here for the fourbar linkage can be used for any number of links in a closed-loop configuration. More complicated linkages may have multiple loops which will lead to more equations to be solved simultaneously and may require an iterative solution.

The Geared Fivebar Linkage

Another example, which can be reduced to two equations in two unknowns, is the **geared fivebar linkage**, which was introduced in Section 2.13 (p. 52) and is shown in Figure 4-11a and program FIVEBAR disk file F04-11.5br. The vector loop for this linkage is shown in Figure 4-11b. It obviously has one more position vector than the fourbar. Its vector loop equation is:

$$\mathbf{R}_2 + \mathbf{R}_3 - \mathbf{R}_4 - \mathbf{R}_5 - \mathbf{R}_1 = 0 \quad (4.23a)$$

Note that the vector senses are again chosen to suit the analyst's desires to have the vector angles defined at a convenient end of the respective link. Equation 4.23b substitutes the complex polar notation for the position vectors in equation 4-23a, using a, b, c, d, f to represent the scalar lengths of the links as shown in Figure 4-11.

$$ae^{j\theta_2} + be^{j\theta_3} - ce^{j\theta_4} - de^{j\theta_5} - fe^{j\theta_1} = 0 \quad (4.23b)$$

Note also that this vector loop equation has three unknown variables in it, namely the angles of links 3, 4, and 5. (The angle of link 2 is the input, or independent, variable, and link 1 is fixed with constant angle). Since a two-dimensional vector equation can only be solved for two unknowns, we will need another equation to solve this system. Because this is a geared fivebar linkage, there exists a relationship between the two geared links, here links 2 and 5. Two factors determine how link 5 behaves with respect to link 2, namely, the **gear ratio** λ and the **phase angle** ϕ . The relationship is:

$$\theta_5 = \lambda\theta_2 + \phi \quad (4.23c)$$

This allows us to express θ_5 in terms of θ_2 in equation 4.23b and reduce the unknowns to two by substituting equation 4.23c into equation 4.23b.

$$ae^{j\theta_2} + be^{j\theta_3} - ce^{j\theta_4} - de^{j(\lambda\theta_2 + \phi)} - fe^{j\theta_1} = 0 \quad (4.24a)$$

Note that the gear ratio λ is the ratio of the diameters of the gears connecting the two links ($\lambda = dia_2 / dia_5$), and the phase angle ϕ is the *initial angle* of link 5 with respect to link 2. When link 2 is at zero degrees, link 5 is at the **phase angle** ϕ . Equation 4.23c defines the relationship between θ_2 and θ_5 . Both λ and ϕ are design parameters selected by the design engineer along with the link lengths. With these parameters defined, the only unknowns left in equation 4.24 are θ_3 and θ_4 .

The behavior of the geared fivebar linkage can be modified by changing the link lengths, the gear ratio, or the phase angle. The phase angle can be changed simply by lifting the gears out of engagement, rotating one gear with respect to the other, and reengaging them. Since links 2 and 5 are rigidly attached to gears 2 and 5, respectively, their relative angular rotations will be changed also. It is this fact that results in different po-

* Waldron and Sreenivasan^[1] report that the common solution methods for position analysis are not general, i.e., are not extendable to n -link mechanisms. Conventional position analysis methods, such as those used here, rely on the presence of a 4-bar loop in the mechanism that can be solved first, followed by a decomposition of the remaining links into a series of dyads. Not all mechanisms contain fourbar loops. (One 8-bar, 1-DOF linkage contains no 4-bar loops.) Even if they do, the 4-bar loop's pivots may not be grounded, requiring that the linkage be inverted to start the solution. Also, if the driving joint is not in the fourbar loop, then interpolation is needed to solve for link positions.

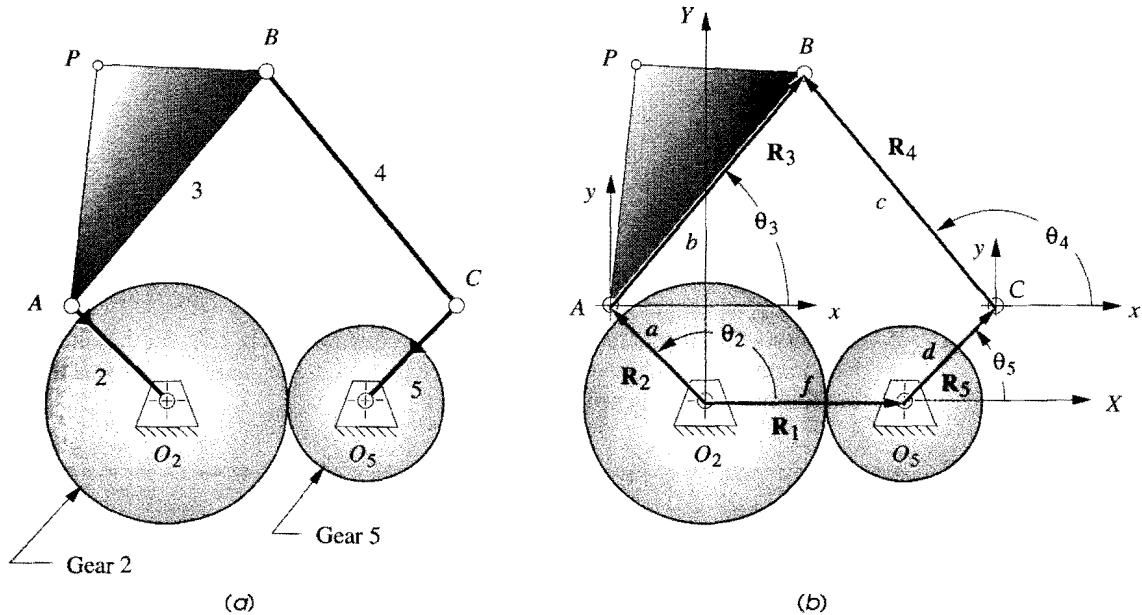


FIGURE 4-11

The geared fivebar linkage and its vector loop

sitions of links 3 and 4 with any change in phase angle. The coupler curve's shapes will also change with variation in any of these parameters as can be seen in Figure 3-20 (p. 109) and in Appendix E.

The procedure for solution of this vector loop equation is the same as that used above for the fourbar linkage:

- 1 Substitute the Euler equivalent (equation 4.4a, p. 155) into each term in the vector loop equation 4.24a.

$$a(\cos\theta_2 + j\sin\theta_2) + b(\cos\theta_3 + j\sin\theta_3) - c(\cos\theta_4 + j\sin\theta_4) - d[\cos(\lambda\theta_2 + \phi) + j\sin(\lambda\theta_2 + \phi)] - f(\cos\theta_1 + j\sin\theta_1) = 0 \quad (4.24b)$$

- 2 Separate the real and imaginary parts of the cartesian form of the vector loop equation.

$$a\cos\theta_2 + b\cos\theta_3 - c\cos\theta_4 - d\cos(\lambda\theta_2 + \phi) - f\cos\theta_1 = 0 \quad (4.24c)$$

$$a\sin\theta_2 + b\sin\theta_3 - c\sin\theta_4 - d\sin(\lambda\theta_2 + \phi) - f\sin\theta_1 = 0 \quad (4.24d)$$

- 3 Rearrange to isolate one unknown (either θ_3 or θ_4) in each scalar equation. Note that θ_1 is zero.

$$b\cos\theta_3 = -a\cos\theta_2 + c\cos\theta_4 + d\cos(\lambda\theta_2 + \phi) + f \quad (4.24e)$$

$$b\sin\theta_3 = -a\sin\theta_2 + c\sin\theta_4 + d\sin(\lambda\theta_2 + \phi) \quad (4.24f)$$

- 4 Square both equations and add them to eliminate one unknown, say θ_3 .

$$\begin{aligned}
 b^2 = & 2c[d\cos(\lambda\theta_2 + \phi) - a\cos\theta_2 + f]\cos\theta_4 \\
 & + 2c[d\sin(\lambda\theta_2 + \phi) - a\sin\theta_2]\sin\theta_4 \\
 & + a^2 + c^2 + d^2 + f^2 - 2af\cos\theta_2 \\
 & - 2d(a\cos\theta_2 - f)\cos(\lambda\theta_2 + \phi) \\
 & - 2ad\sin\theta_2\sin(\lambda\theta_2 + \phi)
 \end{aligned} \quad (4.24g)$$

- 5 Substitute the tangent half-angle identities (equation 4.9, p. 158) for the sine and cosine terms and manipulate the resulting equation in the same way as was done for the fourbar linkage in order to solve for θ_4 .

$$\begin{aligned}
 A &= 2c[d\cos(\lambda\theta_2 + \phi) - a\cos\theta_2 + f] \\
 B &= 2c[d\sin(\lambda\theta_2 + \phi) - a\sin\theta_2] \\
 C &= a^2 - b^2 + c^2 + d^2 + f^2 - 2af\cos\theta_2 \\
 &\quad - 2d(a\cos\theta_2 - f)\cos(\lambda\theta_2 + \phi) \\
 &\quad - 2ad\sin\theta_2\sin(\lambda\theta_2 + \phi) \\
 D &= C - A; \quad E = 2B; \quad F = A + C
 \end{aligned}$$

$$\theta_{4,1,2} = 2\arctan\left(\frac{-E \pm \sqrt{E^2 - 4DF}}{2D}\right) \quad (4.24h)$$

- 6 Repeat steps 3 to 5 for the other unknown angle θ_3 .

$$\begin{aligned}
 G &= 2b[a\cos\theta_2 - d\cos(\lambda\theta_2 + \phi) - f] \\
 H &= 2b[a\sin\theta_2 - d\sin(\lambda\theta_2 + \phi)] \\
 K &= a^2 + b^2 - c^2 + d^2 + f^2 - 2af\cos\theta_2 \\
 &\quad - 2d(a\cos\theta_2 - f)\cos(\lambda\theta_2 + \phi) \\
 &\quad - 2ad\sin\theta_2\sin(\lambda\theta_2 + \phi) \\
 L &= K - G; \quad M = 2H; \quad N = G + K
 \end{aligned}$$

$$\theta_{3,1,2} = 2\arctan\left(\frac{-M \pm \sqrt{M^2 - 4LN}}{2L}\right) \quad (4.24i)$$

Note that these derivation steps are essentially identical to those for the pin-jointed fourbar linkage once θ_2 is substituted for θ_5 using equation 4.23c.

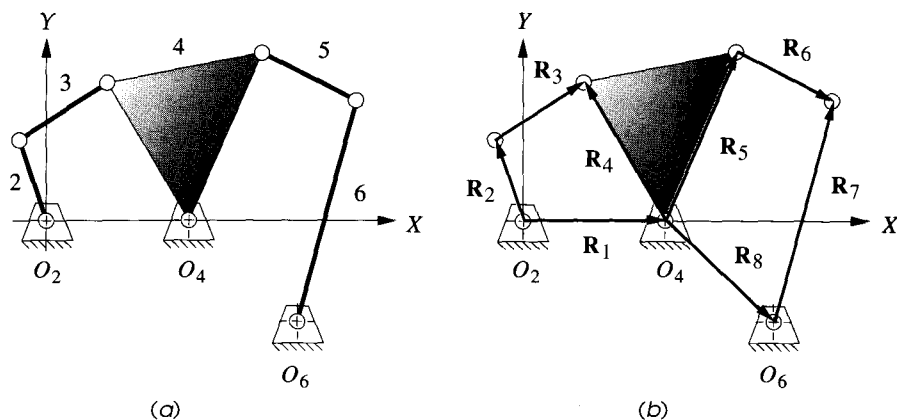


FIGURE 4-12

Watt's sixbar linkage and vector loop

Sixbar Linkages

WATT'S SIXBAR is essentially two fourbar linkages in series, as shown in Figure 4-12a, and can be analyzed as such. Two vector loops are drawn as shown in Figure 4-12b. These vector loop equations can be solved in succession with the results of the first loop applied as input to the second loop. Note that there is a constant angular relationship between vectors R_4 and R_5 within link 4. The solution for the fourbar linkage (equations 4.10 and 4.13, pp. 158 and 159, respectively) is simply applied twice in this case. Depending on the inversion of the Watts linkage being analyzed, there may be two four-link loops or one four-link and one five-link loop. (See Figure 2-14, p. 45.) In either case, if the four-link loop is analyzed first, there will not be more than two unknown link angles to be found at one time.

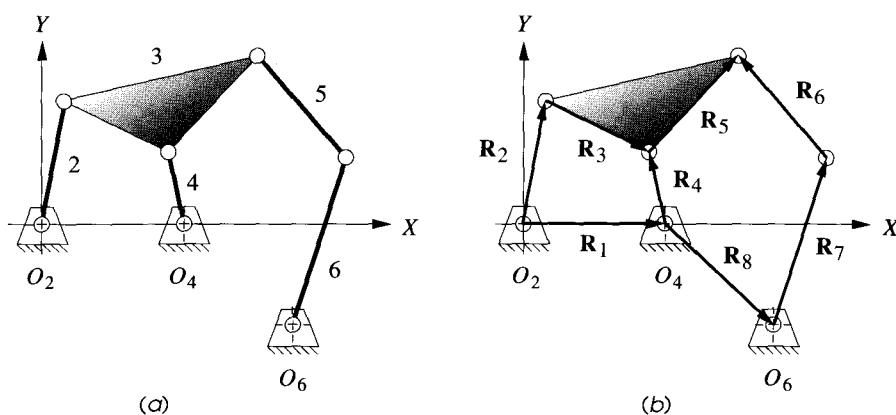


FIGURE 4-13

Stephenson's sixbar linkage and vector loop

STEPHENSON'S SIXBAR is a more complicated mechanism to analyze. Two vector loops can be drawn, but depending on the inversion being analyzed, either one or both loops will have five links* and thus three unknown angles as shown in Figure 4-13a and b (p. 167). However, the two loops will have at least one nonground link in common and so a solution can be found. In the other cases an iterative solution such as a Newton-Raphson method (see Section 4.13) must be used to find the roots of the equations. Program SIXBAR is limited to the inversions which allow a closed-form solution, one of which is shown in Figure 4-13. Program SIXBAR does not do the iterative solution.

4.9 POSITION OF ANY POINT ON A LINKAGE

Once the angles of all the links are found, it is simple and straightforward to define and calculate the position of any point on any link for any input position of the linkage. Figure 4-14 shows a fourbar linkage whose coupler, link 3, is enlarged to contain a coupler point P . The crank and rocker have also been enlarged to show points S and U which might represent the centers of gravity of those links. We want to develop algebraic expressions for the positions of these (or any) points on the links.

To find the position of point S , draw a position vector from the fixed pivot O_2 to point S . This vector $\mathbf{R}_{SO_2}$ makes an angle δ_2 with the vector $\mathbf{R}_{AO_2}$. This angle δ_2 is completely defined by the geometry of link 2 and is constant. The position vector for point S is then:

$$\mathbf{R}_{SO_2} = \mathbf{R}_S = se^{j(\theta_2 + \delta_2)} = s[\cos(\theta_2 + \delta_2) + j\sin(\theta_2 + \delta_2)] \quad (4.25)$$

The position of point U on link 4 is found in the same way, using the angle δ_4 which is a constant angular offset within the link. The expression is:

$$\mathbf{R}_{UO_4} = ue^{j(\theta_4 + \delta_4)} = u[\cos(\theta_4 + \delta_4) + j\sin(\theta_4 + \delta_4)] \quad (4.26)$$

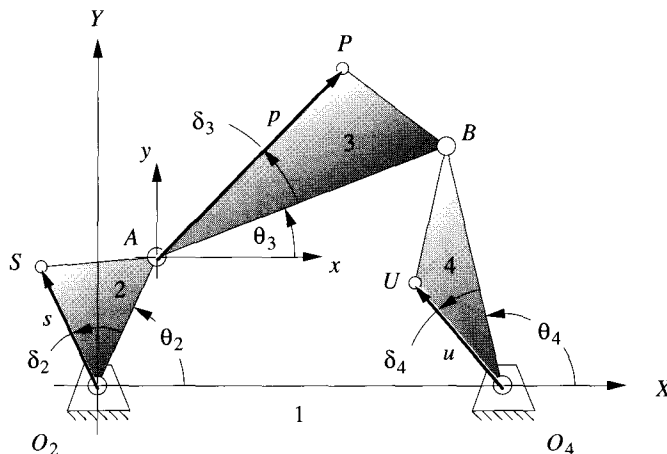


FIGURE 4-14

Positions of points on the links

* See footnote on p. 164.

The position of point P on link 3 can be found from the addition of two position vectors $\mathbf{R}_A$ and $\mathbf{R}_{PA}$. $\mathbf{R}_A$ is already defined from our analysis of the link angles in equation 4.5. $\mathbf{R}_{PA}$ is the relative position of point P with respect to point A . $\mathbf{R}_{PA}$ is defined in the same way as $\mathbf{R}_S$ or $\mathbf{R}_U$, using the internal link offset angle δ_3 and the position angle of link 3, θ_3 .

$$\mathbf{R}_{PA} = p e^{j(\theta_3 + \delta_3)} = p [\cos(\theta_3 + \delta_3) + j \sin(\theta_3 + \delta_3)] \quad (4.27a)$$

$$\mathbf{R}_P = \mathbf{R}_A + \mathbf{R}_{PA} \quad (4.27b)$$

Compare equation 4.27 with equation 4.1. This is also the position difference equation.

4.10 TRANSMISSION ANGLES

The transmission angle was defined in Section 3.3 (p. 81) for a fourbar linkage. That definition is repeated here for your convenience.

The **transmission angle** μ is shown in Figure 3-3a (p. 82) and is defined as *the angle between the output link and the coupler*. It is usually taken as the absolute value of the acute angle of the pair of angles at the intersection of the two links and varies continuously from some minimum to some maximum value as the linkage goes through its range of motion. It is a measure of the quality of force transmission at the joint.*

We will expand that definition here to represent the angle between any two links in a linkage, as a linkage can have many transmission angles. The angle between any output link and the coupler which drives it is a transmission angle. Now that we have developed the analytic expressions for the angles of all the links in a mechanism, it is easy to define the transmission angle algebraically. It is merely the difference between the angles of the two joined links through which we wish to pass some force or velocity. For our fourbar linkage example it will be the difference between θ_3 and θ_4 . By convention we take the absolute value of the difference and force it to be an acute angle.

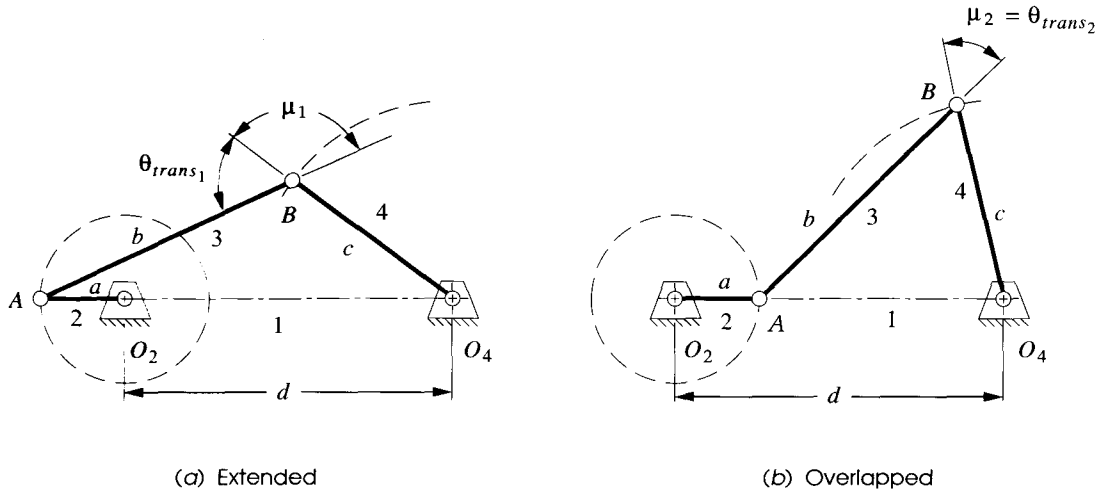
$$\begin{aligned} \theta_{trans} &= |\theta_3 - \theta_4| \\ \text{if } \theta_{trans} &> \frac{\pi}{2} \quad \text{then } \theta_{trans} &= \pi - \theta_{trans} \end{aligned} \quad (4.28)$$

This computation can be done for any joint in a linkage by using the appropriate link angles.

Extreme Values of the Transmission Angle

For a Grashof crank-rocker fourbar linkage the extreme values of the transmission angle will occur when the crank is colinear with the ground link as shown in Figure 4-15. The values of the transmission angle in these positions are easily calculated from the law of cosines since the linkage is then in a triangular configuration. The sides of the two triangles are link 3, link 4, and either the sum or difference of links 1 and 2. One extreme value of the transmission angle occurs when links 1 and 2 are *colinear and nonoverlapping* as shown in Figure 4-15a. The other extreme transmission angle occurs when links 1 and 2 are *colinear and overlapping* as shown in Figure 4-15b. Using notation consistent with Section 4.5 and Figure 4-7 (p. 155) we will label the links:

* The transmission angle has limited application. It only predicts the quality of force or torque transmission if the input and output links are pivoted to ground. If the output force is taken from a floating link (coupler), then the transmission angle is of no value. A different index of merit called the joint force index (JFI) is presented in Chapter 11 which discusses force analysis in linkages. (See Section 11.12 on p. 554.) The JFI is useful for situations in which the output link is floating as well as giving the same kind of information when the output is taken from a link rotating against the ground. However, the JFI requires a complete force analysis of the linkage be done whereas the transmission angle is determined from linkage geometry alone.

**FIGURE 4-15**

Extreme transmission angles in the Grashof fourbar linkage

$$a = \text{link 2}; \quad b = \text{link 3}; \quad c = \text{link 4}; \quad d = \text{link 1}$$

For one extreme transmission angle the cosine law gives

$$\mu_1 = \arccos \left[\frac{b^2 + c^2 - (d + a)^2}{2bc} \right] \quad (4.29a)$$

and for the other extreme transmission angle

$$\mu_2 = \arccos \left[\frac{b^2 + c^2 - (d - a)^2}{2bc} \right] \quad (4.29b)$$

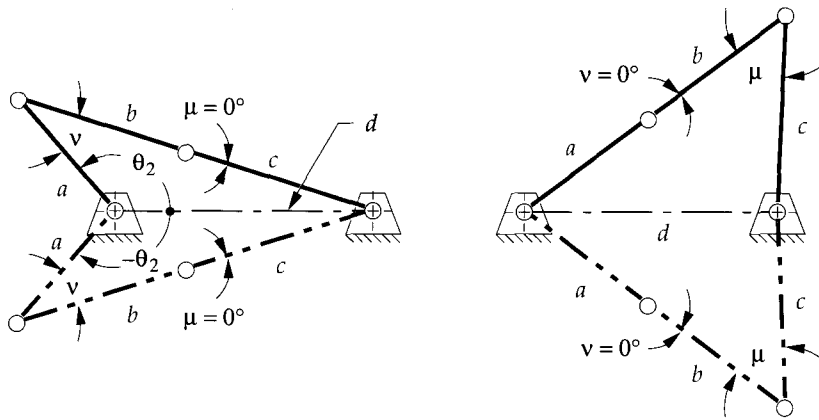
For a **Grashof double-rocker** linkage the transmission angle can vary from 0 to 90 degrees because the coupler can make a full revolution with respect to the other links. For a **non-Grashof triple-rocker** linkage the transmission angle will be zero degrees in the toggle positions which occur when the output rocker c and the coupler b are colinear as shown in Figure 4-16a. In the other toggle positions when input rocker a and coupler b are colinear (Figure 4-16b), the transmission angle can be calculated from the cosine law as:

when

$$v = 0,$$

$$\mu = \arccos \left[\frac{(a + b)^2 + c^2 - d^2}{2c(a + b)} \right] \quad (4.30)$$

This is not the smallest value that the transmission angle μ can have in a triple-rocker, as that will obviously be zero. Of course, when analyzing any linkage, the transmis-



(a) Toggle positions for links b and c

(b) Toggle positions for links a and b

FIGURE 4-16

Non-Grashof triple-rocker linkages in toggle

sion angles can easily be computed and plotted for all positions using equation 4.28. Programs FOURBAR, FIVEBAR, and SIXBAR do this. The student should investigate the variation in transmission angle for the example linkages in those programs. Disk file F04-15.4br can be opened in program FOURBAR to observe that linkage in motion.

4.11 TOGGLE POSITIONS

The input link angles which correspond to the toggle positions (stationary configurations) of the **non-Grashof triple-rocker** can be calculated by the following method, using trigonometry. Figure 4-17 shows a non-Grashof fourbar linkage in a general position. A construction line h has been drawn between points A and O_4 . This divides the quadrilateral loop into two triangles, O_2AO_4 and ABO_4 . Equation 4.31 uses the cosine law to express the transmission angle μ in terms of link lengths and the input link angle θ_2 .

$$h^2 = a^2 + d^2 - 2ad \cos \theta_2$$

also:

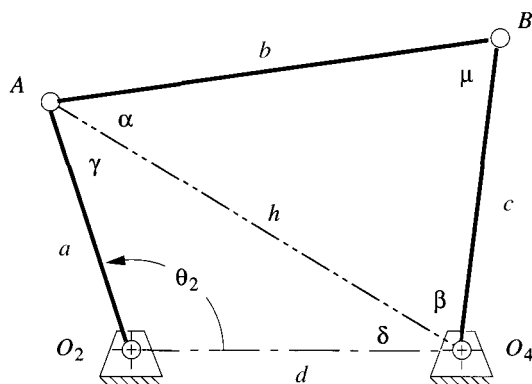
$$h^2 = b^2 + c^2 - 2bc \cos \mu$$

so:

$$a^2 + d^2 - 2ad \cos \theta_2 = b^2 + c^2 - 2bc \cos \mu$$

and:

$$\cos \mu = \frac{b^2 + c^2 - a^2 - d^2}{2bc} + \frac{ad}{bc} \cos \theta_2 \quad (4.31)$$

**FIGURE 4-17**

Finding the crank angle corresponding to the toggle positions

To find the maximum and minimum values of input angle θ_2 , we can differentiate equation 4.31, form the derivative of θ_2 with respect to μ , and set it equal to zero.

$$\frac{d\theta_2}{d\mu} = \frac{bc}{ad} \frac{\sin \mu}{\sin \theta_2} = 0 \quad (4.32)$$

The link lengths a, b, c, d are never zero, so this expression can only be zero when $\sin \mu$ is zero. This will be true when angle μ in Figure 4-17 is either zero or 180° . This is consistent with the definition of toggle given in Section 3.3 (p. 80). If μ is zero or 180° then $\cos \mu$ will be ± 1 . Substituting these two values for $\cos \mu$ into equation 4.31 will give a solution for the value of θ_2 between zero and 180° which corresponds to the toggle position of a triple-rocker linkage when driven from one rocker.

$$\cos \mu = \frac{b^2 + c^2 - a^2 - d^2}{2bc} + \frac{ad}{bc} \cos \theta_2 = \pm 1$$

or :

$$\cos \theta_2 = \frac{a^2 + d^2 - b^2 - c^2}{2ad} \pm \frac{bc}{ad} \quad (4.33)$$

and :

$$\theta_{2_{toggle}} = \arccos \left(\frac{a^2 + d^2 - b^2 - c^2}{2ad} \pm \frac{bc}{ad} \right); \quad 0 \leq \theta_{2_{toggle}} \leq \pi$$

Only one of these $\pm$ cases will produce an argument for the arccosine function which lies between ± 1 . The toggle angle which is in the first or second quadrant can be found from this value. The other toggle angle will then be the negative of the one found, due to the mirror symmetry of the two toggle positions about the ground link as shown in Figure 4-16 (p. 171). Program FOURBAR computes the values of these toggle angles for any non-Grashof linkage.

4.12 CIRCUITS AND BRANCHES IN LINKAGES

In Section 4.5 it was noted that the fourbar linkage position problem has two solutions which correspond to the two circuits of the linkage. This section will explore the topics of circuits and branches in linkages in more detail.

Chase and Mirth[2] define a **circuit** in a linkage as "all possible orientations of the links that can be realized without disconnecting any of the joints" and a **branch** as "a continuous series of positions of the mechanism on a circuit between two stationary configurations The stationary configurations divide a circuit into a series of branches." A linkage may have one or more circuits each of which may contain one or more branches. The number of circuits corresponds to the number of solutions possible from the position equations for the linkage.

Circuit defects are fatal to linkage operation, but branch defects are not. A mechanism that must change circuits to move from one desired position to the other (referred to as a **circuit defect**) is not useful as it cannot do so without disassembly and reassembly. A mechanism that changes branch when moving from one circuit to another (referred to as a **branch defect**) may or may not be usable depending on the designer's intent.

The tailgate linkage shown in Figure 3-2 (p. 81) is an example of a linkage with a deliberate branch defect in its range of motion (actually at the limit of its range of motion). The toggle position (stationary configuration) that it reaches with the tailgate fully open serves to hold it open. But the user can move it out of this stationary configuration by rotating one of the links out of toggle. Folding chairs and tables often use a similar scheme as do fold-down seats in automobiles and station wagons (shooting brakes).

Another example of a common linkage with a branch defect is the slider-crank linkage (crankshaft, connecting rod, piston) used in every piston engine and shown in Figure 13-3 (p. 601). This linkage has two toggle positions (top and bottom dead center) giving it two branches within one revolution of its crank. It works nevertheless because it is carried through these stationary configurations by the angular momentum of the rotating crank and its attached flywheel. One penalty is that the engine must be spun to start it in order to build sufficient momentum to carry it through these toggle positions.

The Watt sixbar linkage can have four circuits, and the Stephenson sixbar can have either four or six circuits depending on which link is driving. Eightbar linkages can have as many as 16 or 18 circuits, not all of which may be real, however[2]

The number of circuits and branches in the fourbar linkage depends on its Grashof condition and the inversion used. A non-Grashof, triple-rocker fourbar linkage has only one circuit but has two branches. All Grashof fourbar linkages have two circuits, but the number of branches per circuit differs with the inversion. The crank-rocker and double-crank have only one branch within each circuit. The double-rocker and rocker-crank have two branches within each circuit. Table 4-1 summarizes these relationships[2]

Any solution for the position of a linkage must take into account the number of possible circuits that it contains. A closed-form solution, if available, will contain all the circuits. An iterative solution such as is described in the next section will only yield the position data for one circuit, and it may not be the one you expect.

TABLE 4-1
Circuits & Branches
In the Fourbar Linkage

Fourbar Linkage Type	Number of Circuits	Branch per Circuit
Non-Grashof triple-rocker	1	2
Grashof* crank-rocker	2	1
Grashof* double-crank	2	1
Grashof* double-rocker	2	2
Grashof* rocker-crank	2	2

* Valid only for non-special case Grashof linkages.

4.13 NEWTON-RAPHSON SOLUTION METHOD

The solution methods for position analysis shown so far in this chapter are all of "closed form," meaning that they provide the solution with a direct, noniterative approach. * In some situations, particularly with multiloop mechanisms, a closed-form solution may not be attainable. Then an alternative approach is needed, and the Newton-Raphson method (sometimes just called Newton's method) provides one that can solve sets of simultaneous nonlinear equations. Any iterative solution method requires that one or more guess values be provided to start the computation. It then uses the guess values to obtain a new solution that may be closer to the correct one. This process is repeated until it converges to a solution close enough to the correct one for practical purposes. However, there is no guarantee that an iterative method will converge at all. It may diverge, taking successive solutions further from the correct one, especially if the initial guess is not sufficiently close to the real solution.

* Kramer [3] states that: "In theory, any nonlinear algebraic system of equations can be manipulated into the form of a single polynomial in one unknown. The roots of this polynomial can then be used to determine all unknowns in the system. However, if the derived polynomial is greater than degree four, factoring and/or some form of iteration are necessary to obtain the roots. In general, systems that have more than a fourth degree polynomial associated with the eliminant of all but one variable must be solved by iteration. However, if factoring of the polynomial into terms of degree four or less is possible, all roots may be found without iteration. Therefore the only truly symbolic solutions are those that can be factored into terms of fourth degree or less. This is the formal definition of a closed form solution."

Though we will need to use the multidimensional (Newton-Raphson version) of Newton's method for these linkage problems, it is easier to understand how the algorithm works by first discussing the one-dimensional Newton's method for finding the roots of a single nonlinear function in one independent variable. Then we will discuss the multidimensional Newton-Raphson method.

One-Dimensional Root-Finding (Newton's Method)

A nonlinear function may have multiple roots, where a root is defined as the intersection of the function with any straight line. Typically the zero axis of the independent variable is the straight line for which we desire the roots. Take, for example, a cubic polynomial which will have three roots, with either one or all three being real.

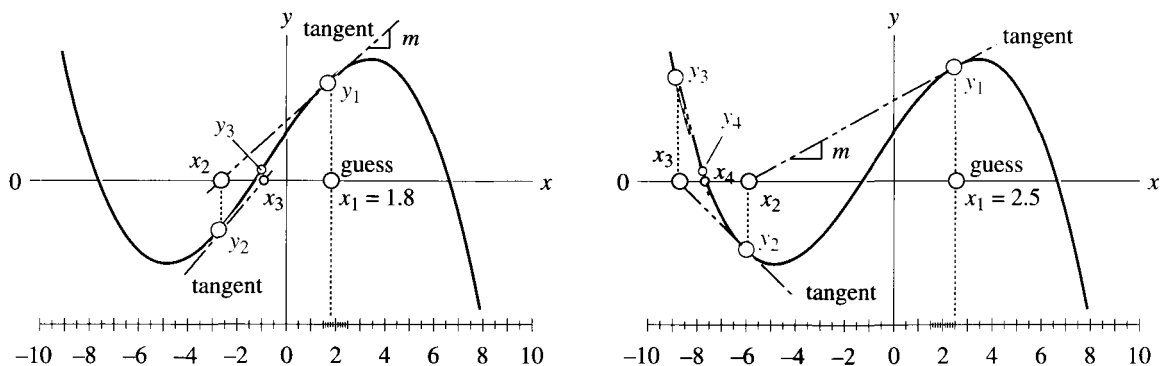
$$y = f(x) = -x^3 - 2x^2 + 50x + 60 \quad (4.34)$$

There is a closed-form solution for the roots of a cubic function t which allows us to calculate in advance that the roots of this particular cubic are all real and are $x = -7.562$, -1.777 , and 6.740 .

Figure 4-18 shows this function plotted over a range of x . In Figure 4-18a, an initial guess value of $x_1 = 1.8$ is chosen. Newton's algorithm evaluates the function for this guess value, finding y_1 . The value of y_1 is compared to a user-selected tolerance (say 0.001) to see if it is close enough to zero to call x_1 the root. If not, then the slope (m) of the function at x_1 , y_1 is calculated either by using an analytic expression for the derivative of the function or by doing a numerical differentiation (less desirable). The equation of the tangent line is then evaluated to find its intercept at x_2 which is used as a new guess value. The above process is repeated, finding y_2 ; testing it against the user selected tolerance; and, if it is too large, calculating another tangent line whose x intercept is used as a new guess value. This process is repeated until the value of the function y_i at the latest x_i is close enough to zero to satisfy the user.

t Viète's method from "De Emendatione" by Francois Viète (1615) as described in reference [4].

The Newton's algorithm described above can be expressed algebraically (in pseudocode) as shown in equation 4.35. The function for which the roots are sought is $f(x)$, and its derivative is $f'(x)$. The slope m of the tangent line is equal to $f'(x)$ at the current point x_i , y_i .



(a) A guess of $x = 1.8$ converges to the root at $x = -1.777$ (b) A guess of $x = 2.5$ converges to the root at $x = -7.562$

FIGURE 4-18

Newton-Raphson method of solution for roots of nonlinear functions

- | | | |
|--------|---|--------|
| step 1 | $y_i = f(x_i)$ | |
| step 2 | IF $y_i \leq \text{tolerance}$ THEN STOP | |
| step 3 | $m = f'(x_i)$ | |
| step 4 | $x_{i+1} = x_i - \frac{y_i}{m}$ | |
| step 5 | $y_{i+1} = f(x_{i+1})$ | |
| step 6 | IF $y_{i+1} \leq \text{tolerance}$ THEN STOP | |
| | ELSE $x_i = x_{i+1} : y_i = y_{i+1} : \text{GOTO step 1}$ | (4.35) |

If the initial guess value is close to a root, this algorithm will converge rapidly to the solution. However, it is quite sensitive to the initial guess value. Figure 4-18b shows the result of a slight change in the initial guess from $x_1 = 1.8$ to $x_1 = 2.5$. With this slightly different guess it converges to another root. Note also that if we choose an initial guess of $x_1 = 3.579$ which corresponds to a local maximum of this function, the tangent line will be horizontal and will not intersect the x axis at all. The method fails in this situation. Can you suggest a value of x_1 that would cause it to converge to the root at $x = 6.74$?

So this method has its drawbacks. It may fail to converge. It may behave chaotically.* It is sensitive to the guess value. It also is incapable of distinguishing between multiple circuits in a linkage. The circuit solution it finds is dependent on the initial guess. It requires that the function be differentiable, and the derivative as well as the function must be evaluated at every step. Nevertheless, it is the method of choice for functions whose derivatives can be efficiently evaluated and which are continuous in the region of the root. Furthermore, it is about the only choice for systems of nonlinear equations.

* Kramer[3] points out that "the Newton Raphson algorithm can exhibit chaotic behavior when there are multiple solutions to kinematic constraint equations Newton Raphson has no mechanism for distinguishing between the two solutions" (circuits). He does an experiment with just two links, exactly analogous to finding the angles of the coupler and rocker in the fourbar linkage position problem, and finds that the initial guess values need to be quite close to the desired solution (one of the two possible circuits) to avoid divergence or chaotic oscillation between the two solutions.

Multidimensional Root-Finding (Newton-Raphson Method)

The one-dimensional Newton's method is easily extended to multiple, simultaneous, nonlinear equation sets and is then called the Newton-Raphson method. First, let's generalize the expression developed for the one-dimensional case in step 4 of equation 4.35. Refer also to Figure 4-18 (p. 175).

$$\begin{aligned}
 & x_{i+1} = x_i - \frac{y_i}{m} \quad \text{or} \quad m(x_{i+1} - x_i) = -y_i \\
 \text{but :} \quad & y_i = f(x_i) \quad m = f'(x_i) \quad x_{i+1} - x_i = \Delta x \\
 \\
 & \text{substituting :} \quad f'(x_i) \cdot \Delta x = -f(x_i) \quad (4.36)
 \end{aligned}$$

Here a Δx term is introduced which will approach zero as the solution converges. The Δx term rather than y_i will be tested against a selected tolerance in this case. Note that this form of the equation avoids the division operation which is acceptable in a scalar equation but impossible with a matrix equation.

A multidimensional problem will have a set of equations of the form

$$\begin{bmatrix} f_1(x_1, x_2, x_3, \dots, x_n) \\ f_2(x_1, x_2, x_3, \dots, x_n) \\ \vdots \\ f_n(x_1, x_2, x_3, \dots, x_n) \end{bmatrix} = \mathbf{B} \quad (4.37)$$

where the set of equations constitutes a vector, here called **B**.

Partial derivatives are required to obtain the slope terms

$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} = \mathbf{A} \quad (4.38)$$

which form the *jacobian matrix* of the system, here called **A**.

The error terms are also a vector, here called **X**.

$$\begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix} = \mathbf{X} \quad (4.39)$$

Equation 4.36 then becomes a matrix equation for the multidimensional case.

$$\mathbf{A} \mathbf{X} = -\mathbf{B} \quad (4.40)$$

Equation 4.40 can be solved for $\mathbf{X}$ either by matrix inversion or by gaussian elimination. The values of the elements of $\mathbf{A}$ and $\mathbf{B}$ are calculable for any assumed (guess) values of the variables. A criterion for convergence can be taken as the sum of the error vector $\mathbf{X}$ at each iteration where the sum approaches zero at a root.

Let's set up this Newton-Raphson solution for the fourbar linkage.

Newton-Raphson Solution for the Fourbar Linkage

The vector loop equation of the fourbar linkage, separated into its real (equation 4.6a) and imaginary (equation 4.6b) parts provides the set of functions that define the two unknown link angles θ_3 and θ_4 . The link lengths, a , b , c , d , and the input angle θ_2 are given.

$$f_1 = a \cos \theta_2 + b \cos \theta_3 - c \cos \theta_4 - d = 0 \quad (4.41a)$$

$$f_2 = a \sin \theta_2 + b \sin \theta_3 - c \sin \theta_4 = 0$$

$$\mathbf{B} = \begin{bmatrix} a \cos \theta_2 + b \cos \theta_3 - c \cos \theta_4 - d \\ a \sin \theta_2 + b \sin \theta_3 - c \sin \theta_4 \end{bmatrix} \quad (4.41b)$$

The error vector is:

$$\mathbf{X} = \begin{bmatrix} \Delta \theta_3 \\ \Delta \theta_4 \end{bmatrix} \quad (4.42)$$

The partial derivatives are:

$$\mathbf{A} = \begin{bmatrix} \frac{\partial f_1}{\partial \theta_3} & \frac{\partial f_1}{\partial \theta_4} \\ \frac{\partial f_2}{\partial \theta_3} & \frac{\partial f_2}{\partial \theta_4} \end{bmatrix} = \begin{bmatrix} -b \sin \theta_3 & c \sin \theta_4 \\ b \cos \theta_3 & -c \cos \theta_4 \end{bmatrix} \quad (4.43)$$

This matrix is known as the **jacobian** of the system, and, in addition to its usefulness in this solution method, it also tells something about the solvability of the system. The system of equations for position, velocity, and acceleration (in all of which the jacobian appears) can only be solved if the value of the determinant of the jacobian is nonzero.

Substituting equations 4.41b, 4.42, and 4.43 into equation 4.40 gives:

$$\begin{bmatrix} -b \sin \theta_3 & c \sin \theta_4 \\ b \cos \theta_3 & -c \cos \theta_4 \end{bmatrix} \begin{bmatrix} \Delta \theta_3 \\ \Delta \theta_4 \end{bmatrix} = - \begin{bmatrix} a \cos \theta_2 + b \cos \theta_3 - c \cos \theta_4 - d \\ a \sin \theta_2 + b \sin \theta_3 - c \sin \theta_4 \end{bmatrix} \quad (4.44)$$

To solve this matrix equation, guess values will have to be provided for θ_3 and θ_4 and the two equations then solved simultaneously for $\Delta \theta_3$ and $\Delta \theta_4$. For a larger system of equations, a matrix reduction algorithm will need to be used. For this simple system in two unknowns, the two equations can be solved by combination and reduction. The test described above which compares the sum of the values of $\Delta \theta_3$ and $\Delta \theta_4$ to a selected tolerance must be applied after each iteration to determine if a root has been found.

Equation Solvers

Some commercially available equation solver software packages include the ability to do a Newton-Raphson iterative solution on sets of nonlinear simultaneous equations. *TKSolver** and *Mathcaat* are examples. *TKSolver* automatically invokes its Newton-Raphson solver when it cannot directly solve the presented equation set, provided that enough guess values have been supplied for the unknowns. These equation solver tools are quite convenient in that the user need only supply the equations for the system in "raw" form such as equation 4.4la. It is not necessary to arrange them into the Newton-Raphson algorithm as shown in the previous section. Lacking such a commercial equation solver, you will have to write your own computer code to program the solution as described above. Reference [5] is a useful aid in this regard. The CD-ROM included with this text contains example *TKSolver* files for the solution of this fourbar position problem as well as others.

4.14 REFERENCES

- 1 Waldron, K. J., and S. V. Sreenivasan. (1996). "A Study of the Solvability of the Position Problem for Multi-Circuit Mechanisms by Way of Example of the Double Butterfly Linkage." *Journal of Mechanical Design*, 118(3), p. 390.
- 2 Chase, T. R., and J. A. Mirth. (1993). "Circuits and Branches of Single-Degree-of-Freedom Planar Linkages." *Journal of Mechanical Design*, 115, p. 223.
- 3 Kramer, G. (1992). *Solving Geometric Constraint Systems: A Case Study in Kinematics*. MIT Press: Cambridge, pp. 155-158.
- 4 Press, W. H., et al. (1986). *Numerical Recipes: The Art of Scientific Computing*. Cambridge University Press: Cambridge, pp. 145-146.
- 5 Ibid, pp. 254-273.

4.15 PROBLEMS

- 4-1 A position vector is defined as having a length equal to your height in inches (or centimeters). The tangent of its angle is defined as your weight in pounds (or kilograms) divided by your age in years. Calculate the data for this vector and:
 - a. Draw the position vector to scale on cartesian axes.
 - b. Write an expression for the position vector using unit vector notation.
 - c. Write an expression for the position vector using complex number notation, in both polar and cartesian forms.
- 4-2 A particle is traveling along an arc of 6.5-in radius. The arc center is at the origin of a coordinate system. When the particle is at position *A*, its position vector makes a 45° angle with the X axis. At position *B*, its vector makes a 75° angle with the X axis. Draw this system to some convenient scale and:
 - a. Write an expression for the particle's position vector in position *A* using complex number notation, in both polar and cartesian forms.
 - b. Write an expression for the particle's position vector in position *B* using complex number notation, in both polar and cartesian forms.
 - c. Write a vector equation for the position difference between points *B* and *A*. Substitute the complex number notation for the vectors in this equation and solve for the position difference numerically.
 - d. Check the result of part c with a graphical method.

* Universal Technical Systems, 1220 Rock St. Rockford, IL 61101, USA. (800) 435-7887

t Mathsoft, 201 Broadway, Cambridge, MA 02139 (800) 628-4223

- 4-3 Repeat problem 4-2 considering points A and B to represent separate particles, and find their relative position.
- 4-4 Repeat Problem 4-2 with the particle's path defined as being along the line $y = -2x + 10$.
- 4-5 Repeat Problem 4-3 with the path of the particle defined as being along the curve $y = -2x^2 - 2x + 10$.
- *4-6 The link lengths and the value of θ_2 for some fourbar linkages are defined in Table P4-1. The linkage configuration and terminology are shown in Figure P4-1. For the rows assigned, draw the linkage to scale and graphically find all possible solutions (both open and crossed) for angles θ_3 and θ_4 . Determine the Grashof condition.
- *†4-7 Repeat Problem 4-6 except solve by the vector loop method.
- 4-8 Expand equation 4.7b and prove that it reduces to equation 4.7c (p. 157).
- *4-9 The link lengths and the value of θ_2 and offset for some fourbar slider-crank linkages are defined in Table P4-2. The linkage configuration and terminology are shown in Figure P4-2. For the rows assigned, draw the linkage to scale and graphically find all possible solutions (both open and crossed) for angle θ_3 and slider position d .
- *†4-10 Repeat Problem 4-9 except solve by the vector loop method.
- *4-11 The link lengths and the value of θ_2 and γ for some inverted fourbar slider-crank linkages are defined in Table P4-3. The linkage configuration and terminology are shown in Figure P4-3. For the rows assigned, draw the linkage to scale and graphically find both open and crossed solutions for angles θ_3 and θ_4 and vector $\mathbf{R}_B$.
- *†4-12 Repeat Problem 4-11 except solve by the vector loop method.
- *†4-13 Find the transmission angles of the linkages in the assigned rows in Table P4-1.

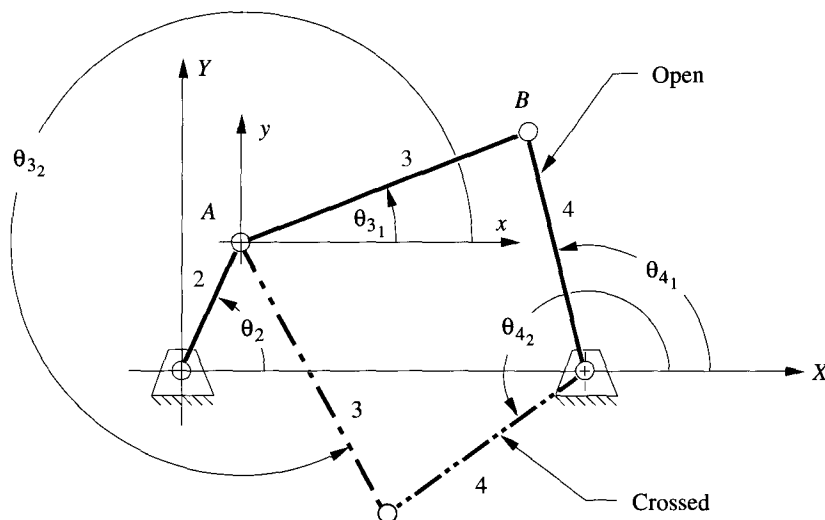


FIGURE P4-1

Problems 4-6 to 4-7. General configuration and terminology for the fourbar linkage

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program FOURBAR, SLIDER, or SIXBAR.

TABLE P4-1 Data for Problems 4-6, 4-7 and 4-13 to 4-15

Row	Link 1	Link 2	Link 3	Link 4	θ_2
a	6	2	7	9	30
b	7	9	3	8	85
c	3	10	6	8	45
d	8	5	7	6	25
e	8	5	8	6	75
f	5	8	8	9	15
g	6	8	8	9	25
h	20	10	10	10	50
i	4	5	2	5	80
j	20	10	5	10	33
k	4	6	10	7	88
l	9	7	10	7	60
m	9	7	11	8	50
n	9	7	11	6	120

- *†4-14 Find the minimum and maximum values of the transmission angle for all the Grashof crank-rocker linkages in Table P4-1.
- *†4-15 Find the input angles corresponding to the toggle positions of the non-Grashof linkages in Table P4-1. (For this problem, ignore the values of θ_2 given in the table.)
- *4-16 The link lengths, gear ratio (λ), phase angle (ϕ), and the value of θ_2 for some geared fivebar linkages are defined in Table P4-4. The linkage configuration and terminology are shown in Figure P4-4. For the rows assigned, draw the linkage to scale and graphically find all possible solutions for angles θ_3 and θ_4 .
- *†4-17 Repeat Problem 4-16 except solve by the vector loop method.

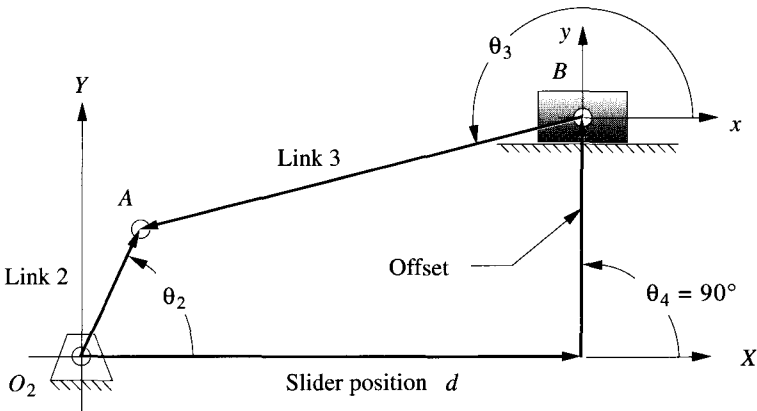


FIGURE P4-2
Problems 4-9 to 4-10 Open configuration and terminology for a fourbar slider-crank linkage

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program FOURBAR, SLIDER, or SIXBAR.

TABLE P4-2 Data for Problems 4-9 to 4-10

Row	Link 2	Link 3	Offset	θ_2
a	1.4	4	1	45
b	2	6	-3	60
c	3	8	2	-30
d	3.5	10	1	120
e	5	20	-5	225
f	3	13	0	100
g	7	25	10	330

TABLE P4-3 Data for Problems 4-11 to 4-12

Row	Link 1	Link 2	Link 4	γ	θ_2
a	6	2	4	90	30
b	7	9	3	75	85
c	3	10	6	45	45
d	8	5	3	60	25
e	8	4	2	30	75
f	5	8	8	90	150

4-18 Scale Figure P4-5 to obtain dimensions for the following problems each of which refers to the part of the figure with the same letter. Reference all angular measurements to a line of centers.

- a. Find the angular displacement of link CD when link AB rotates clockwise from the position shown to horizontal. What is the smallest transmission angle between

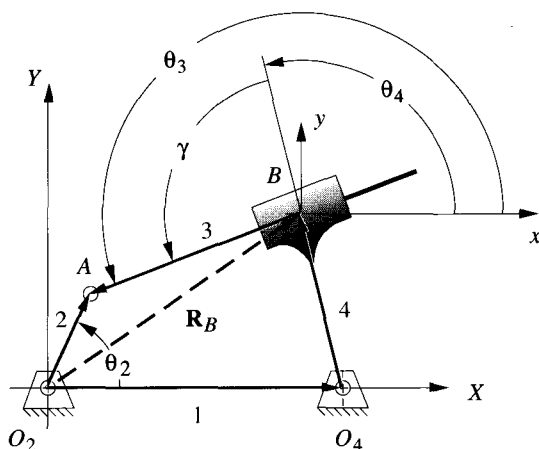
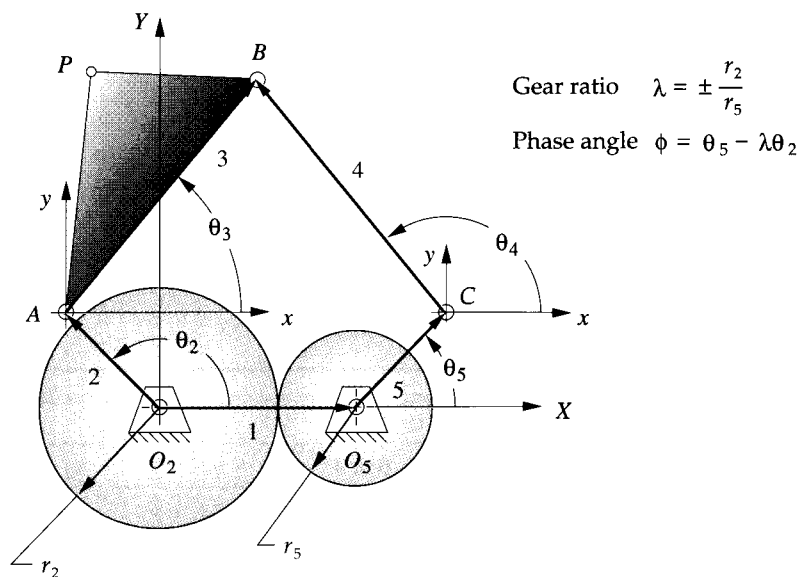


FIGURE P4-3

Problems 4-11 to 4-12. Terminology for inversion #3 of the fourbar slider-crank linkage

**FIGURE P4-4**

Problems 4-16 to 4-17. Open configuration and terminology for the geared fivebar linkage

those two positions? Find the toggle positions of this linkage in terms of the angle of link AB .

- Find and plot the angular position of links BC and CD and the transmission angle as a function of the angle of wheel W as it rotates through one revolution.
- Find and plot the position of any one piston as a function of the angle of crank AB as it rotates through one revolution. Once one piston's motion is defined, find the motions of the other two pistons and their phase relationship to the first piston.
- Find the total angular displacement of link BC as link AB makes a complete revolution. Determine the total stroke of the box at point E as link AB makes a complete revolution.

TABLE P4-4 Data for Problems 4-16 to 4-17

Row	Link 1	Link 2	Link 3	Link 4	Link 5	λ	ϕ	θ_2
a	6	1	7	9	4	2	30	60
b	6	5	7	8	4	-2.5	60	30
c	3	5	7	8	4	-0.5	0	45
d	4	5	7	8	4	-1	120	75
e	5	9	11	8	8	3.2	-50	-39
f	10	2	7	5	3	1.5	30	120
g	15	7	9	11	4	2.5	-90	75
h	12	8	7	9	4	-2.5	60	55
i	9	7	8	9	4	-4	120	100

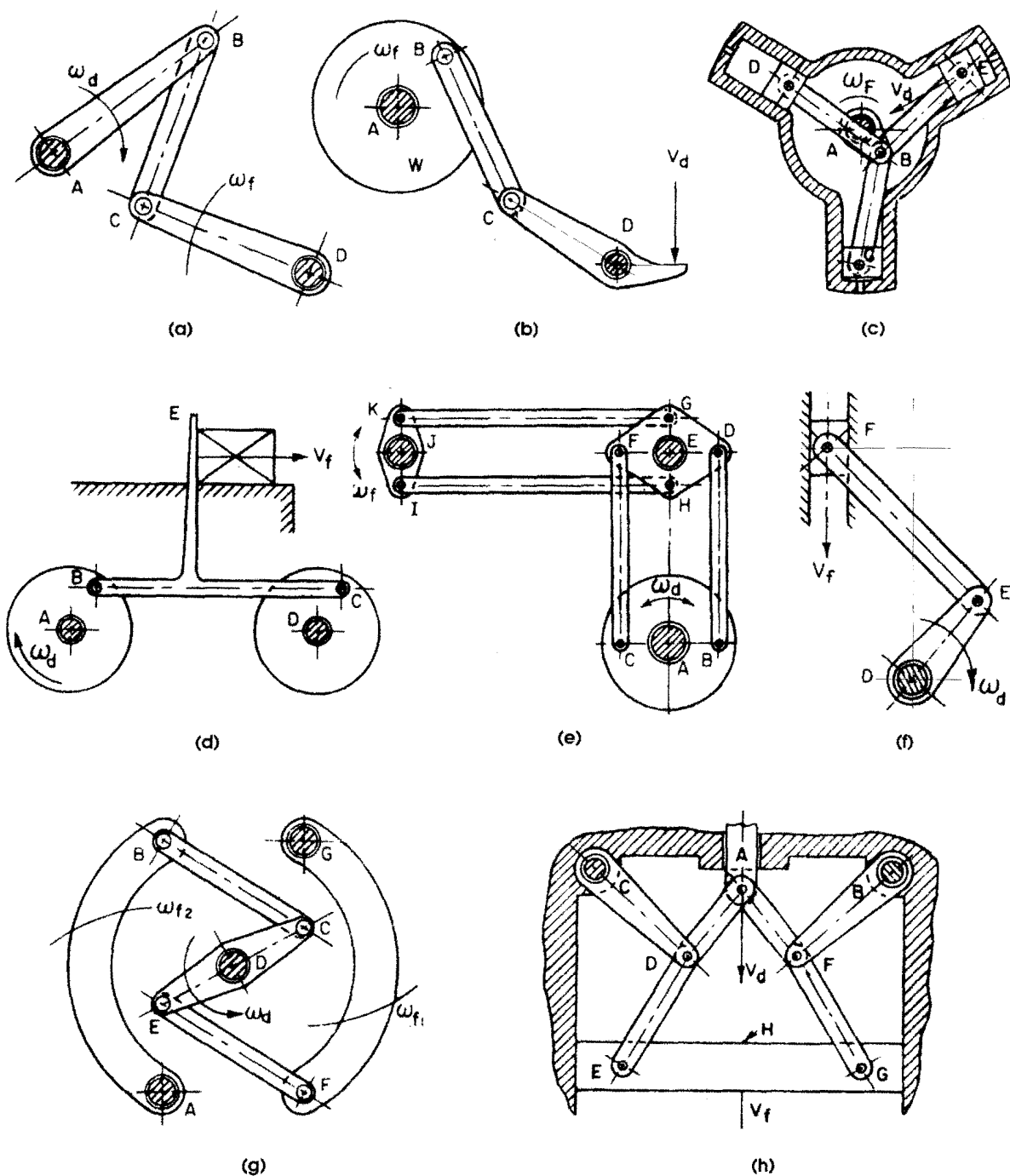
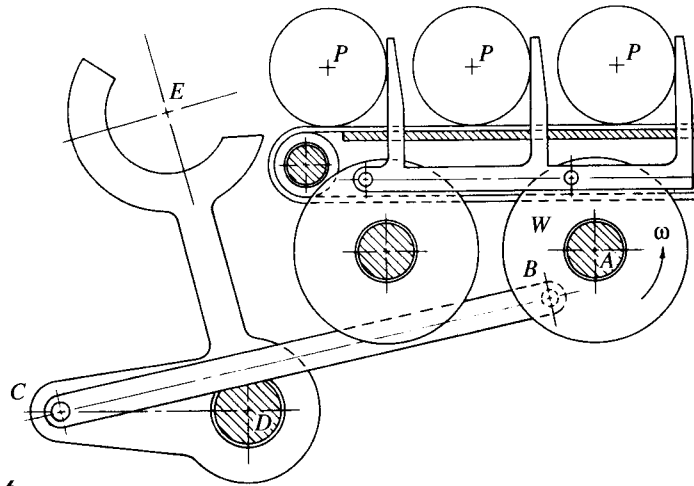


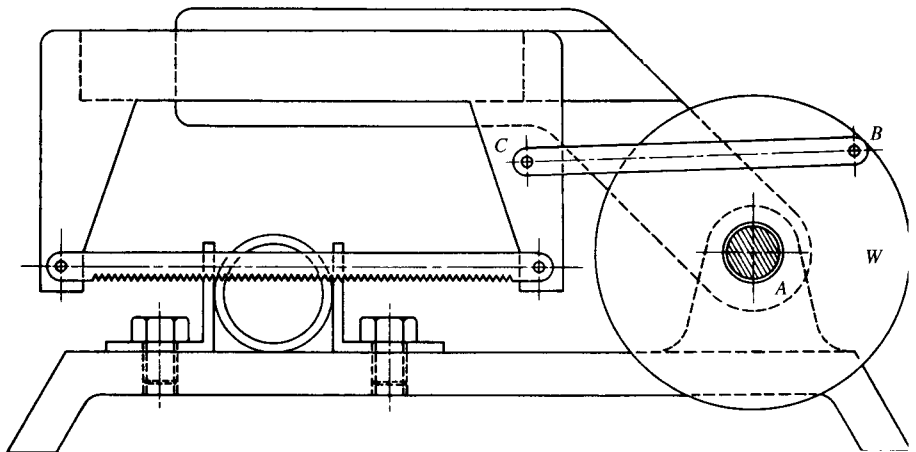
FIGURE P4-5

Mechanisms for Problem 4-18. Adapted from P. H. Hill and W. P. Rule. (1960). *Mechanisms: Analysis and Design*, with permission.

**FIGURE P4-6**

Pusher and pick-and-place mechanism for Problem 4-19. Adapted from P. H. Hill and W. P. Rule. (1960). *Mechanisms: Analysis and Design*, with permission.

- e. Determine the ratio of angular displacement between link AB and JK as a function of angular displacement of input crank JK . Plot the transmission angle at point G for one revolution of crank JK . Comment on the behavior of this linkage.
- f. Find and plot the displacement of piston F and the angular displacement of link EF as a function of the angular displacement of crank DE .
- g. Find and plot the angular displacement of link AB versus the angle of input link DC as it is rotated from the position shown to a vertical position. Find the toggle positions of this linkage in terms of the angle of link DC .
- h. Find point G 's maximum displacement vertically downward from the position shown. What will the angle of input link BF be at that position?

**FIGURE P4-7**

Power hacksaw for Problem 4-20. Adapted from P. H. Hill and W. P. Rule. (1960). *Mechanisms: Analysis and Design*, with permission.

- †4-19 For one revolution of driving wheel W of the pusher and pick-and-place mechanism in Figure P4-6, find the horizontal stroke of the pushers for the portion of their motion where their tips are above the top belt. Express the stroke as a percentage of the crank length AB . What portion of a revolution of the driver crank AB does this stroke correspond to? Also find the total angular displacement of link DE over one revolution of wheel W . Scale the drawing for all necessary dimensions.
- †4-20 For one revolution of driving wheel W of the hacksaw mechanism on the cutting stroke depicted in Figure P4-7, find and plot the horizontal stroke of the saw blade as a function of the angle of input crank AB .
- *†4-21 For the linkage in Figure P4-8, find its limit (toggle) positions in terms of the angle of link O_2A referenced to the line of centers O_2O_4 when driven from link O_2A . Then calculate and plot the xy coordinates of coupler point P between those limits, referenced to the line of centers O_2O_4 .
- †4-22 For the walking beam mechanism of Figure P4-9, calculate and plot the x and y components of the position of the coupler point P for one complete revolution of the crank O_2A . Hint: Calculate them first with respect to the ground link O_2O_4 and then transform them into the global XY coordinate system (i.e., horizontal and vertical in the figure). Scale the figure for any additional information needed.
- *†4-23 For the linkage in Figure P4-10, calculate and plot the angular displacement of links 3 and 4 and the path coordinates of point P with respect to the angle of the input crank O_2A for one revolution.
- †4-24 For the linkage in Figure P4-11, calculate and plot the angular displacement of links 3 and 4 with respect to the angle of the input crank O_2A for one revolution.
- *†4-25 For the linkage in Figure P4-12, find its limit (toggle) positions in terms of the angle of link O_2A referenced to the line of centers O_2O_4 when driven from link O_2A . Then calculate and plot the angular displacement of links 3 and 4 and the path coordinates of point P with respect to the angle of the input crank O_2A over its possible range of motion referenced to the line of centers O_2O_4 .
- *†4-26 For the linkage in Figure P4-13, find its limit (toggle) positions in terms of the angle of link O_2A referenced to the line of centers O_2O_4 when driven from link O_2A . Then calculate and plot the angular displacement of links 3 and 4 and the path coordinates

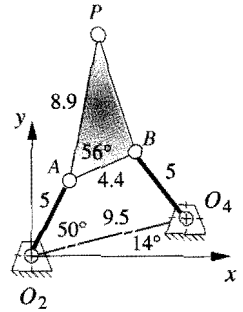


FIGURE P4-8

Problem 4-21

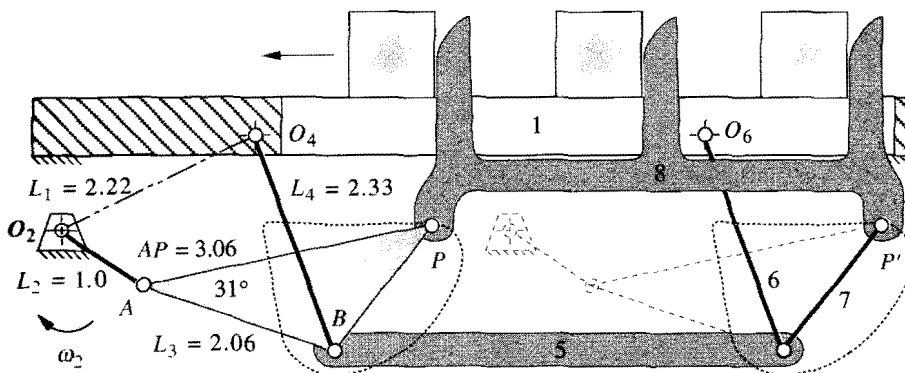


FIGURE P4-9

Problem 4-22. Straight-line walking beam eightbar transport mechanism

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program *FOURBAR*, *SLIDER*, or *SIXBAR*.

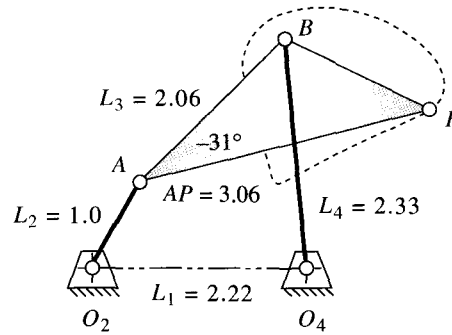


FIGURE P4-10

Problem 4-23

of point P between those limits, with respect to the angle of the input crank O_2A over its possible range of motion referenced to the line of centers O_2O_4 .

†4-27 For the linkage in Figure P4-13, find its limit (toggle) positions in terms of the angle of link O_4B referenced to the line of centers O_4O_2 when driven from link O_4B . Then calculate and plot the angular displacement of links 2 and 3 and the path coordinates of point P between those limits, with respect to the angle of the input crank O_4B over its possible range of motion referenced to the line of centers O_4O_2 .

†4-28 For the rocker-crank linkage in Figure P4-14, find the maximum angular displacement possible for the treadle link (to which force F is applied). Determine the toggle positions. How does this work? Explain why the grinding wheel is able to fully rotate despite the presence of toggle positions when driven from the treadle. How would you get it started if it was in a toggle position?

*†4-29 For the linkage in Figure P4-15, find its limit (toggle) positions in terms of the angle of link O_2A referenced to the line of centers O_2O_4 when driven from link O_2A . Then calculate and plot the angular displacement of links 3 and 4 and the path coordinates of point P between those limits, with respect to the angle of the input crank O_2A over its possible range of motion referenced to the line of centers O_2O_4 .

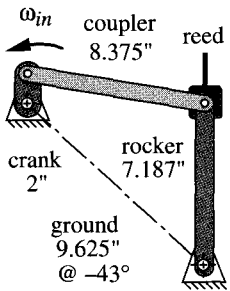


FIGURE P4-11

Problem 4-24

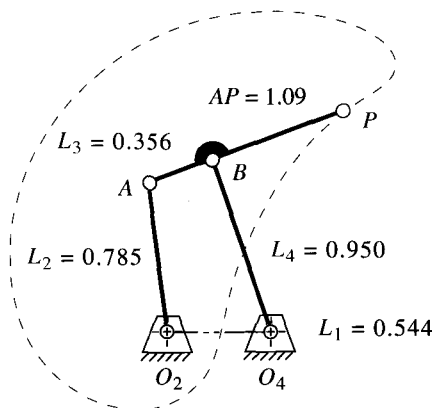


FIGURE P4-12

Problem 4-25

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program *FOURBAR*, *SLIDER*, or *SIXBAR*.

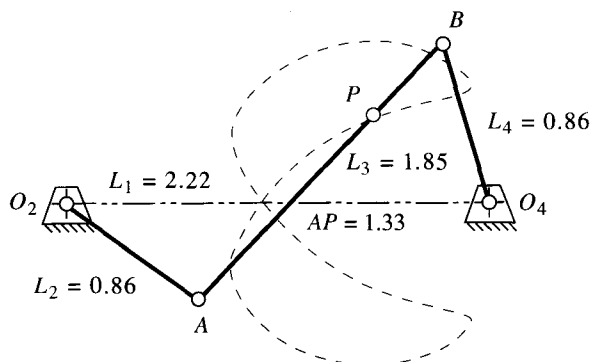


FIGURE P4-13

Problems 4-26 to 4-27

- *†4-30 For the linkage in Figure P4-15, find its limit (toggle) positions in terms of the angle of link O_4B referenced to the line of centers O_4O_2 when driven from link O_4B . Then calculate and plot the angular displacement of links 3 and 4 and the path coordinates of point P between those limits, with respect to the angle of the input crank O_4B over its possible range of motion referenced to the line of centers O_4O_2 .
- †4-31 Write a computer program (or use an equation solver such as *Mathcad*, *Matlab*, or *TKSolver*) to find the roots of $y = 9x^2 + 50x - 40$. Hint: Plot the function to determine good guess values.
- †4-32 Write a computer program (or use an equation solver such as *Mathcad*, *Matlab*, or *TKSolver*) to find the roots of $y = -x^3 - 4x^2 + 80x - 40$. Hint: Plot the function to determine good guess values.
- †4-33 Figure 4-18 (p.175) plots the cubic function from equation 4.34. Write a computer program (or use an equation solver such as *Mathcad*, *Matlab*, or *TKSolver* to solve the matrix equation) to investigate the behavior of the Newton-Raphson algorithm as the initial guess value is varied from $x = 1.8$ to 2.5 in steps of 0.1 . Determine the guess value at which the convergence switches roots. Explain this root-switching phenomenon based on your observations from this exercise.

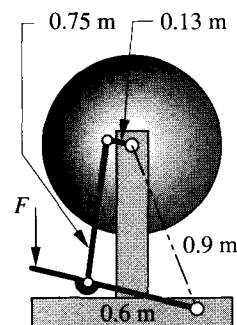


FIGURE P4-14

Problem 4-28

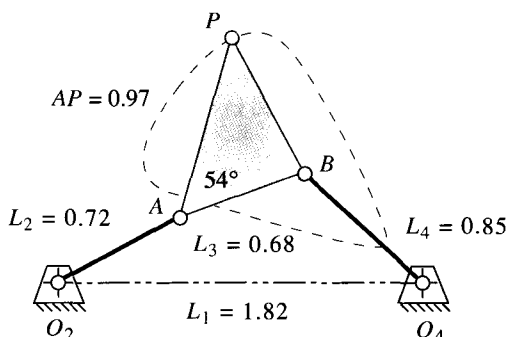
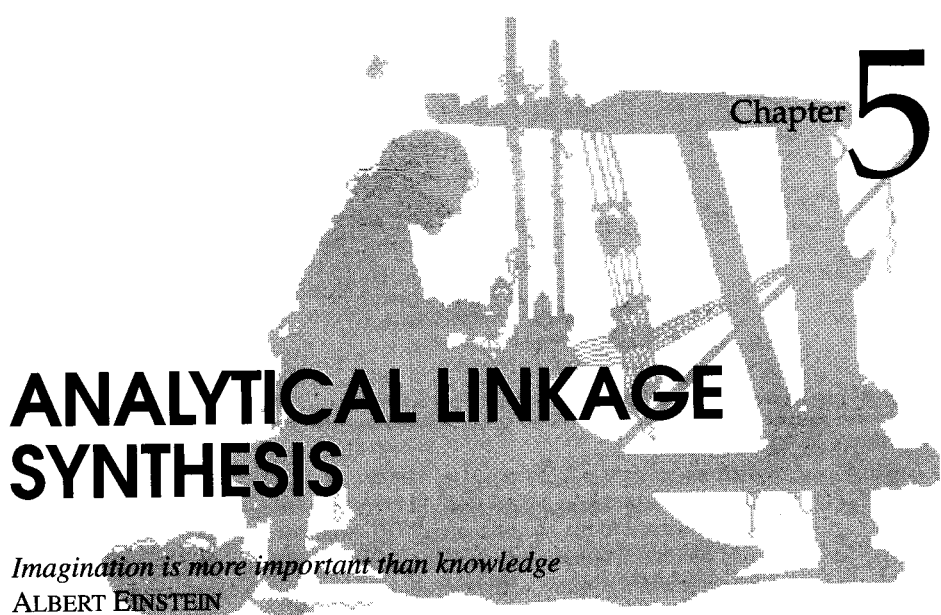


FIGURE P4-15

Problems 4-29 to 4-30

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program FOURBAR, SLIDER, or SIXBAR.



ANALYTICAL LINKAGE SYNTHESIS

Imagination is more important than knowledge

ALBERT EINSTEIN

5.0 INTRODUCTION

With the fundamentals of position analysis established, we can now use these techniques to synthesize linkages for specified output positions analytically. The synthesis techniques presented in Chapter 3 were strictly graphical and somewhat intuitive. The analytical synthesis procedure is algebraic rather than graphical and is less intuitive. However, its algebraic nature makes it quite suitable for computerization. These analytical synthesis methods were originated by Sandor[1] and further developed by his students, Erdman,[2] Kaufman,[3] and Loerch et al.[4,5]

5.1 TYPES OF KINEMATIC SYNTHESIS

Erdman and Sandor[6] define three types of kinematic synthesis, function, path, and motion generation, which were discussed in Section 3.2. Brief definitions are repeated here for your convenience.

FUNCTION GENERATION is defined as *the correlation of an input function with an output function in a mechanism*. Typically, a double-rocker or crank-rocker is the result, with pure rotation input and pure rotation output. A slider-crank linkage can be a function generator as well, driven from either end, i.e., rotation in and translation out or vice versa.

PATH GENERATION is defined as *the control of a point in the plane such that it follows some prescribed path*. This is typically accomplished with a fourbar crank-rocker or double-rocker, wherein a point on the coupler traces the desired output path. No

attempt is made in path generation to control the orientation of the link which contains the point of interest. The coupler curve is made to pass through a set of desired output points. However, it is common for the timing of the arrival of the coupler point at particular locations along the path to be defined. This case is called *path generation with prescribed timing* and is analogous to function generation in that a particular output function is specified.

MOTION GENERATION is defined as *the control of a line in the plane such that it assumes some sequential set of prescribed positions*. Here orientation of the link containing the line is important. This is typically accomplished with a fourbar crank-rocker or double-rocker, wherein a point on the coupler traces the desired output path and the linkage also controls the angular orientation of the coupler link containing the output line of interest.

5.2 PRECISION POINTS

The *points, or positions, prescribed for successive locations of the output (coupler or rocker) link in the plane* are generally referred to as precision points or precision positions. The number of precision points which can be synthesized is limited by the number of equations available for solution. The fourbar linkage can be synthesized by closed-form methods for up to five precision points for motion or path generation with prescribed timing (coupler output) and up to seven points for function generation (rocker output). Synthesis for two or three precision points is relatively straightforward, and each of these cases can be reduced to a system of linear simultaneous equations easily solved on a calculator. The four or more position synthesis problems involve the solution of nonlinear, simultaneous equation systems, and so are more complicated to solve, requiring a computer.

Note that these analytical synthesis procedures provide a solution which will be able to "be at" the specified precision points, but no guarantee is provided regarding the linkage's behavior between those precision points. It is possible that the resulting linkage will be incapable of moving from one precision point to another due to the presence of a toggle position or other constraint. This situation is actually no different than that of the graphical synthesis cases in Chapter 3, wherein there was also the possibility of a toggle position between design points. In fact, these analytical synthesis methods are just an alternate way to solve the same multi position synthesis problems. One should still build a simple cardboard model of the synthesized linkage to observe its behavior and check for the presence of problems, even if the synthesis was performed by an esoteric analytical method.

5.3 TWO-POSITION MOTION GENERATION BY ANALYTICAL SYNTHESIS

Figure 5-1 shows a fourbar linkage in one position with a coupler point located at a first precision position P_1' . It also indicates a second precision position (point P_2) to be achieved by the rotation of the input rocker, link 2, through an as yet unspecified angle Z . Note also that the angle of the coupler link 3 at each of the precision positions is defined by the angles of the position vectors Z_1 and Z_2 . The angle ϕ corresponds to the

The problem statement is:

Design a fourbar linkage which will move a line on its coupler link such that a point P on that line will be first at P_1 and later at P_2 and will also rotate the line through an angle α_2 between those two precision positions. Find the lengths and angles of the four links and the coupler link dimensions A_1P_1 and B_1P_1 as shown in Figure 5-1.

The two-position analytical motion synthesis procedure is as follows:

Define the two desired precision positions in the plane with respect to an arbitrarily chosen global coordinate system XY using position vectors $\mathbf{R}_1$ and $\mathbf{R}_2$ as shown in Figure 5-1a. The change in angle α_2 of vector $\mathbf{Z}$ is the rotation required of the coupler link. Note that the position difference vector $\mathbf{P}_{21}$ defines the displacement of the output motion of point P and is defined as:

$$\mathbf{P}_{21} = \mathbf{R}_2 - \mathbf{R}_1 \quad (5.1)$$

The dyad $\mathbf{W}_1\mathbf{Z}_1$ defines the left half of the linkage. The dyad $\mathbf{U}_1\mathbf{S}_1$ defines the right half of the linkage. Note that $\mathbf{Z}_1$ and $\mathbf{S}_1$ are both embedded in the rigid coupler (link 3), and both of these vectors will undergo the same rotation through angle α_2 from position 1 to position 2. The pin-to-pin length and angle of link 3 (vector $\mathbf{V}_1$) is defined in terms of vectors $\mathbf{Z}_1$ and $\mathbf{S}_1$.

$$\mathbf{V}_1 = \mathbf{Z}_1 - \mathbf{S}_1 \quad (5.2a)$$

The ground link 1 is also definable in terms of the two dyads.

$$\mathbf{G}_1 = \mathbf{W}_1 + \mathbf{V}_1 - \mathbf{U}_1 \quad (5.2b)$$

Thus if we can define the two dyads $\mathbf{W}_1, \mathbf{Z}_1$, and $\mathbf{U}_1, \mathbf{S}_1$, we will have defined a linkage that meets the problem specifications.

We will first solve for the left side of the linkage (vectors $\mathbf{W}_1$ and $\mathbf{Z}_1$) and later use the same procedure to solve for the right side (vectors $\mathbf{U}_1$ and $\mathbf{S}_1$). To solve for $\mathbf{W}_1$ and $\mathbf{Z}_1$ we need only write a vector loop equation around the loop which includes both positions P_1 and P_2 for the left-side dyad. We will go clockwise around the loop, starting with $\mathbf{W}_2$.

$$\mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{P}_{21} - \mathbf{Z}_1 - \mathbf{W}_1 = 0 \quad (5.3)$$

Now substitute the complex number equivalents for the vectors.

$$we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} - p_{21}e^{j\delta_2} - ze^{j\phi} - we^{j\theta} = 0 \quad (5.4)$$

The sums of angles in the exponents can be rewritten as products of terms.

$$we^{j\theta}e^{j\beta_2} + ze^{j\phi}e^{j\alpha_2} - p_{21}e^{j\delta_2} - ze^{j\phi} - we^{j\theta} = 0 \quad (5.5a)$$

Simplifying and rearranging:

$$we^{j\theta}(e^{j\beta_2} - 1) + ze^{j\phi}(e^{j\alpha_2} - 1) = p_{21}e^{j\delta_2} \quad (5.5b)$$

Note that the lengths of vectors $\mathbf{W}_1$ and $\mathbf{W}_2$ are the same magnitude w because they represent the same rigid link in two different positions. The same can be said about vectors $\mathbf{Z}_1$ and $\mathbf{Z}_2$ whose common magnitude is z .

Equations 5.5 are vector equations, each of which contains two scalar equations and so can be solved for two unknowns. The two scalar equations can be revealed by substituting Euler's identity (equation 4.4a, p. 155) and separating the real and imaginary terms as was done in Section 4.5 (p. 152).

real part:

$$\begin{aligned} [w \cos \theta](\cos \beta_2 - 1) - [w \sin \theta] \sin \beta_2 \\ + [z \cos \phi](\cos \alpha_2 - 1) - [z \sin \phi] \sin \alpha_2 = p_{21} \cos \delta_2 \end{aligned} \quad (5.6a)$$

imaginary part (with complex operator j divided out):

$$\begin{aligned} [w \sin \theta](\cos \beta_2 - 1) + [w \cos \theta] \sin \beta_2 \\ + [z \sin \phi](\cos \alpha_2 - 1) + [z \cos \phi] \sin \alpha_2 = p_{21} \sin \delta_2 \end{aligned} \quad (5.6b)$$

There are eight variables in these two equations: w , θ , β_2 , z , ϕ , α_2 , p_{21} , and δ_2 . We can only solve for two. Three of the eight are defined in the problem statement, namely α_2 , p_{21} , and δ_2 . Of the remaining five, w , θ , β_2 , z , ϕ , we are forced to choose three as "free choices" (assumed values) in order to solve for the other two.

One strategy is to assume values for the three angles, θ , β_2 , ϕ , on the premise that we may want to specify the orientation θ , ϕ of the two link vectors $\mathbf{W}_1$ and $\mathbf{Z}_1$ to suit packaging constraints, and also specify the angular excursion β_2 of link 2 to suit some driving constraint. This choice also has the advantage of leading to a set of equations which are linear in the unknowns and are thus easy to solve. For this solution, the equations can be simplified by setting the assumed and specified terms to be equal to some constants.

In equations 5.6a, let:

$$\begin{aligned} A &= \cos \theta (\cos \beta_2 - 1) - \sin \theta \sin \beta_2 \\ B &= \cos \phi (\cos \alpha_2 - 1) - \sin \phi \sin \alpha_2 \\ C &= p_{21} \cos \delta_2 \end{aligned} \quad (5.7a)$$

and in equations 5.6b let:

$$\begin{aligned} D &= \sin \theta (\cos \beta_2 - 1) + \cos \theta \sin \beta_2 \\ E &= \sin \phi (\cos \alpha_2 - 1) + \cos \phi \sin \alpha_2 \\ F &= p_{21} \sin \delta_2 \end{aligned} \quad (5.7b)$$

then:

$$\begin{aligned} Aw + Bz &= C \\ Dw + Ez &= F \end{aligned} \quad (5.7c)$$

and solving simultaneously,

$$w = \frac{CE - BF}{AE - BD}; \quad z = \frac{AF - CD}{AE - BD} \quad (5.7d)$$

A second strategy is to assume a length z and angle ϕ for vector $\mathbf{Z}_1$ and the angular excursion β_2 of link 2 and then solve for the vector $\mathbf{W}_1$. This is a commonly used approach. Note that the terms in square brackets in each of equations 5.6 are respectively the x and y components of the vectors $\mathbf{W}_1$ and $\mathbf{Z}_1$.

$$\begin{aligned} W_{1x} &= w \cos \theta; & Z_{1x} &= z \cos \phi \\ W_{1y} &= w \sin \theta; & Z_{1y} &= z \sin \phi \end{aligned} \quad (5.8a)$$

Substituting in equation 5.6,

$$\begin{aligned} W_{1x}(\cos \beta_2 - 1) - W_{1y} \sin \beta_2 \\ + Z_{1x}(\cos \alpha_2 - 1) - Z_{1y} \sin \alpha_2 &= p_{21} \cos \delta_2 \\ W_{1y}(\cos \beta_2 - 1) + W_{1x} \sin \beta_2 \\ + Z_{1y}(\cos \alpha_2 - 1) + Z_{1x} \sin \alpha_2 &= p_{21} \sin \delta_2 \end{aligned} \quad (5.8b)$$

Z_{1x} and Z_{1y} are known from equation 5.8a with z and ϕ assumed as free choices. To further simplify the expression, combine other known terms as:

$$\begin{aligned} A &= \cos \beta_2 - 1; & B &= \sin \beta_2; & C &= \cos \alpha_2 - 1 \\ D &= \sin \alpha_2; & E &= p_{21} \cos \delta_2; & F &= p_{21} \sin \delta_2 \end{aligned} \quad (5.8c)$$

substituting,

$$\begin{aligned} AW_{1x} - BW_{1y} + CZ_{1x} - DZ_{1y} &= E \\ AW_{1y} + BW_{1x} + CZ_{1y} + DZ_{1x} &= F \end{aligned} \quad (5.8d)$$

and the solution is:

$$\begin{aligned} W_{1x} &= \frac{A(-CZ_{1x} + DZ_{1y} + E) + B(-CZ_{1y} - DZ_{1x} + F)}{-2A} \\ W_{1y} &= \frac{A(-CZ_{1y} - DZ_{1x} + F) + B(CZ_{1x} - DZ_{1y} - E)}{-2A} \end{aligned} \quad (5.8e)$$

Either of these strategies results in the definition of a left dyad $\mathbf{W}_1\mathbf{Z}_1$ and its pivot locations which will provide the motion generation specified.

We must repeat the process for the right-hand dyad, $\mathbf{U}_1\mathbf{S}_1$. Figure 5-2 highlights the two positions $\mathbf{U}_1\mathbf{S}_1$ and $\mathbf{U}_2\mathbf{S}_2$ of the right dyad. Vector $\mathbf{U}_1$ is initially at angle σ and moves through angle γ_2 from position 1 to 2. Vector $\mathbf{S}_1$ is initially at angle ψ . Note that the rotation of vector $\mathbf{S}$ from $\mathbf{S}_1$ to $\mathbf{S}_2$ is through the same angle α_2 as vector $\mathbf{Z}$, since they are in the same link. A vector loop equation similar to equation 5.3 can be written for this dyad.

$$\mathbf{U}_2 + \mathbf{S}_2 - \mathbf{P}_{21} - \mathbf{S}_1 - \mathbf{U}_1 = 0 \quad (5.9a)$$

Rewrite in complex variable form and collect terms.

$$ue^{j\sigma}(e^{j\gamma_2} - 1) + se^{j\psi}(e^{j\alpha_2} - 1) = p_{21}e^{j\delta_2} \quad (5.9b)$$

When this is expanded and the proper angles substituted, the x and y component equations become:

real part:

$$\begin{aligned} u \cos \sigma (\cos \gamma_2 - 1) - u \sin \sigma \sin \gamma_2 \\ + s \cos \psi (\cos \alpha_2 - 1) - s \sin \psi \sin \alpha_2 = p_{21} \cos \delta_2 \end{aligned} \quad (5.10a)$$

imaginary part (with complex operator j divided out):

$$\begin{aligned} u \sin \sigma (\cos \gamma_2 - 1) + u \cos \sigma \sin \gamma_2 \\ + s \sin \psi (\cos \alpha_2 - 1) + s \cos \psi \sin \alpha_2 = p_{21} \sin \delta_2 \end{aligned} \quad (5.10b)$$

Compare equations 5.10 to equations 5.6.

The same first strategy can be applied to equations 5.10 as was used for equations 5.6 to solve for the magnitudes of vectors $\mathbf{U}$ and $\mathbf{S}$, assuming values for angles σ , ψ , and γ_2 . The quantities p_{21} , δ_2 , and α_2 are defined from the problem statement as before.

In equations 5.10a let:

$$\begin{aligned} A &= \cos \sigma (\cos \gamma_2 - 1) - \sin \sigma \sin \gamma_2 \\ B &= \cos \psi (\cos \alpha_2 - 1) - \sin \psi \sin \alpha_2 \\ C &= p_{21} \cos \delta_2 \end{aligned} \quad (5.11a)$$

and in equations 5.10b let:

$$\begin{aligned} D &= \sin \sigma (\cos \gamma_2 - 1) + \cos \sigma \sin \gamma_2 \\ E &= \sin \psi (\cos \alpha_2 - 1) + \cos \psi \sin \alpha_2 \\ F &= p_{21} \sin \delta_2 \end{aligned} \quad (5.11b)$$

then:

$$\begin{aligned} Au + Bs &= C \\ Du + Es &= F \end{aligned} \quad (5.11c)$$

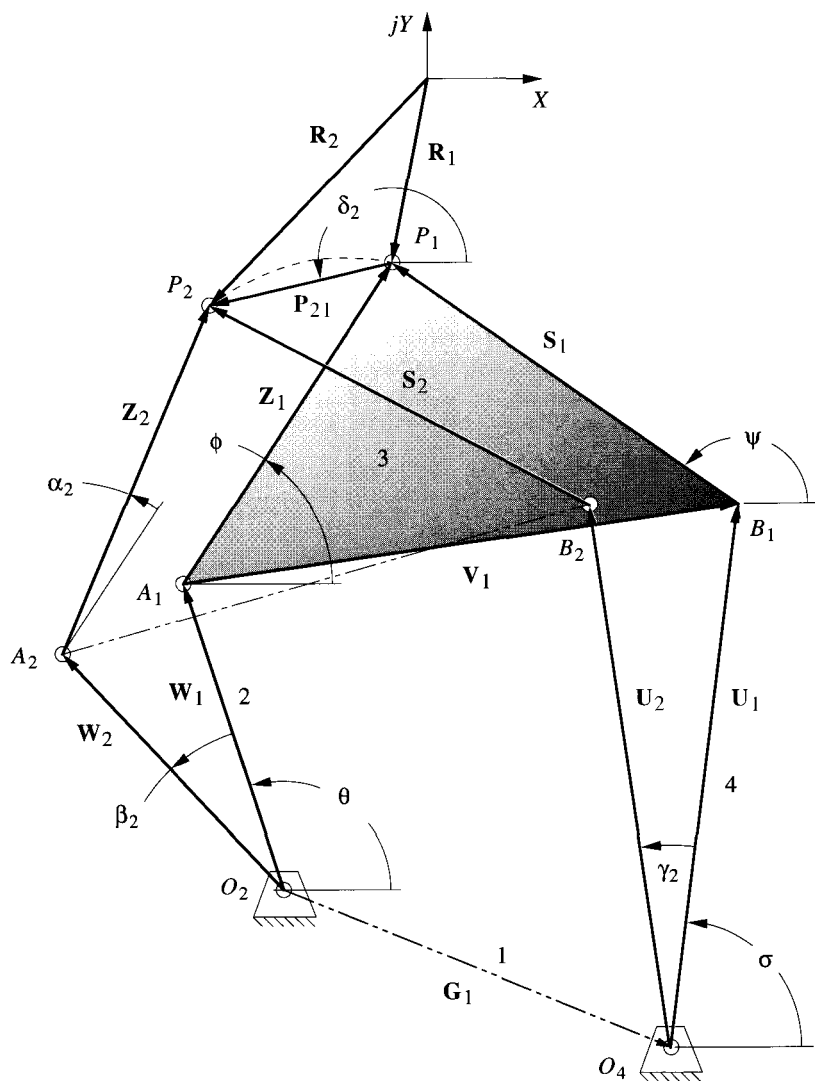
and solving simultaneously,

$$u = \frac{CE - BF}{AE - BD}; \quad s = \frac{AF - CD}{AE - BD} \quad (5.11d)$$

If the second strategy is used, assuming angle γ_2 and the magnitude and direction of vector $\mathbf{S}_1$ (which will define link 3), the result will be:

$$\begin{aligned} U_{1x} &= u \cos \sigma; & S_{1x} &= s \cos \psi \\ U_{1y} &= u \sin \sigma; & S_{1y} &= s \sin \psi \end{aligned} \quad (5.12a)$$

Substitute in equation 5.10:

**FIGURE 5-2**

Right-side dyad shown in two positions

$$\begin{aligned}
 &U_{1x} (\cos \gamma_2 - 1) - U_{1y} \sin \gamma_2 \\
 &\quad + S_{1x} (\cos \alpha_2 - 1) - S_{1y} \sin \alpha_2 = p_{21} \cos \delta_2 \\
 &U_{1y} (\cos \gamma_2 - 1) + U_{1x} \sin \gamma_2 \\
 &\quad + S_{1y} (\cos \alpha_2 - 1) + S_{1x} \sin \alpha_2 = p_{21} \sin \delta_2
 \end{aligned} \tag{5.12b}$$

$$\begin{aligned} \text{Let :} \quad A &= \cos \gamma_2 - 1; & B &= \sin \gamma_2; & C &= \cos \alpha_2 - 1 \\ D &= \sin \alpha_2; & E &= p_{21} \cos \delta_2; & F &= p_{21} \sin \delta_2 \end{aligned} \quad (5.12c)$$

Substitute in equation 5.12b,

$$\begin{aligned} AU_{1x} - BU_{1y} + CS_{1x} - DS_{1y} &= E \\ AU_{1y} + BU_{1x} + CS_{1y} + DS_{1x} &= F \end{aligned} \quad (5.12d)$$

and the solution is:

$$\begin{aligned} U_{1x} &= \frac{A(-CS_{1x} + DS_{1y} + E) + B(-CS_{1y} - DS_{1x} + F)}{-2A} \\ U_{1y} &= \frac{A(-CS_{1y} - DS_{1x} + F) + B(CS_{1x} - DS_{1y} - E)}{-2A} \end{aligned} \quad (5.12e)$$

Note that there are infinities of possible solutions to this problem because we may choose any set of values for the three free choices of variables in this two-position case. Technically there is an infinity of solutions for each free choice. Three choices then give infinity cubed solutions! But since infinity is defined as a number larger than the largest number you can think of, infinity cubed is not any more impressively large than just plain infinity. While not strictly correct mathematically, we will, for simplicity, refer to all of these cases as having “an infinity of solutions,” regardless of the power to which infinity may be raised as a result of the derivation. There are plenty of solutions to pick from, at any rate. *Unfortunately, not all will work.* Some will have CBO defects such as toggle positions between the precision points. Others will have poor transmission angles or poor pivot locations or overlapped links. Design judgment is still most important in selecting the assumed values for your free choices. Despite their name you must pay for those “free choices” later. Make a model!

5.4 COMPARISON OF ANALYTICAL AND GRAPHICAL TWO-POSITION SYNTHESIS

Note that in the **graphical solution** to this two-position synthesis problem (in Example 3-3 and Figure 3-6, p. 87), we also had to make *three free choices* to solve the problem. The identical two-position synthesis problem from Figure 3-6 is reproduced in Figure 5-3. The approach taken in Example 3-3 used the two points *A* and *B* as the attachments for the moving pivots. Figure 5-3a shows the graphical construction used to find the fixed pivots O_2 and O_4 . For the analytical solution we will use those points *A* and *B* as the joints of the two dyads **WZ** and **US**. These dyads meet at point *P*, which is the precision point. The relative position vector $\mathbf{P}_{21}$ defines the displacement of the precision point.

Note that in the graphical solution, we implicitly defined the left dyad vector **Z** by locating attachment points *A* and *B* on link 3 as shown in Figure 5-3a. This defined the two variables, z and ϕ . We also implicitly chose the value of w by selecting an arbitrary location for pivot O_2 on the perpendicular bisector. When that third choice was made,

the remaining two unknowns, angles β_2 and θ , were solved for graphically at the same time, because the geometric construction was in fact a graphical “computation” for the solution of the two simultaneous equations 5.8.

The graphical and analytical methods represent two alternate solutions to the same problem. All of these problems can be solved both analytically and graphically. One method can provide a good check for the other. We will now solve this problem analytically and correlate the results with the graphical solution from Chapter 3.

EXAMPLE 5-1

Two-Position Analytical Motion Synthesis.

Problem: Design a fourbar linkage to move the link APB shown from position $A_1P_1B_1$ to $A_2P_2B_2$.

Solution: (see Figure 5-3)

- 1 Draw the link APB in its two desired positions, $A_1P_1B_1$ and $A_2P_2B_2$, to scale in the plane as shown.
- 2 Measure or calculate the values of the magnitude and angle of vector $\mathbf{P}_{21}$, namely, p_{21} and δ_2 . In this example they are:

$$p_{21} = 2.416; \quad \delta_2 = 165.2^\circ$$

- 3 Measure or calculate the value of the change in angle, α_2 , of vector $\mathbf{Z}$ from position 1 to position 2. In this example it is:

$$\alpha_2 = 43.3^\circ$$

- 4 The three values in steps 2 and 3 are the only ones defined in the problem statement. We must assume three additional “free choices” to solve the problem. Method two (see equations 5.8) chooses the length z and angle ϕ of vector $\mathbf{Z}$ and β_2 , the change in angle of vector $\mathbf{W}$. In order to obtain the same solution as the graphical method produced in Figure 5-3a (from the infinities of solutions available), we will choose those values consistent with the graphical solution.

$$z = 1.298; \quad \phi = 26.5^\circ; \quad \beta_2 = 38.4^\circ$$

- 5 Substitute these six values in equations 5.8 and obtain:

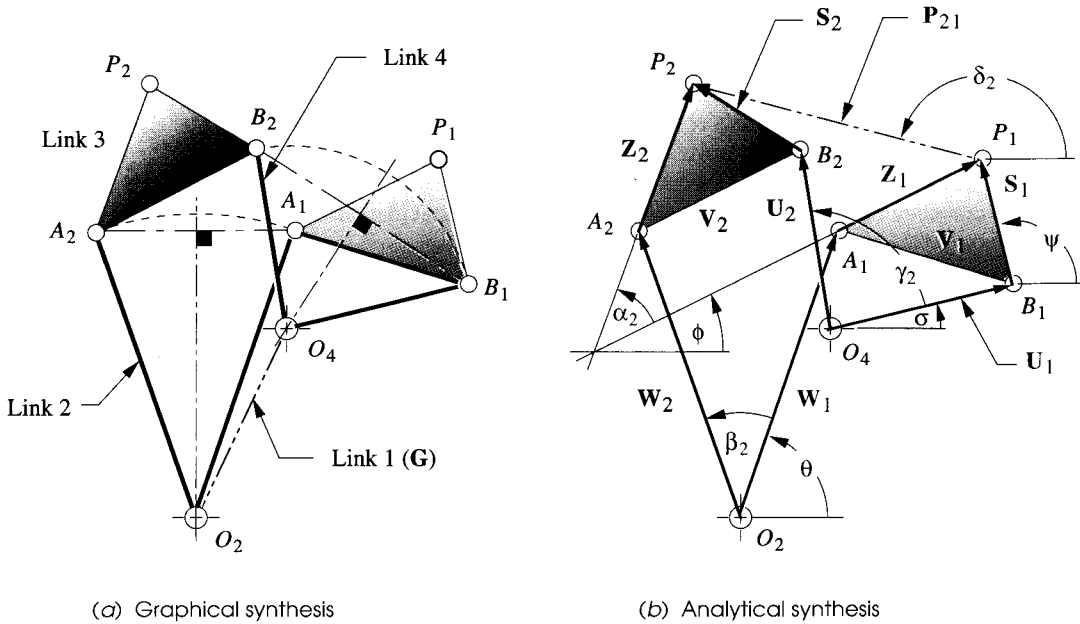
$$w = 2.467 \quad \theta = 71.6^\circ$$

- 6 Compare these to the graphical solution;

$$w = 2.48 \quad \theta = 71^\circ$$

which is a reasonable match given the graphical accuracy. This vector $\mathbf{W}_1$ is link 2 of the fourbar.

- 7 Repeat the procedure for the link-4 side of the linkage. The free choices will now be:



(a) Graphical synthesis

(b) Analytical synthesis

FIGURE 5-3

Two-position motion synthesis with coupler output

$$s = 1.035; \quad \psi = 104.1^\circ; \quad \gamma_2 = 85.6^\circ$$

- 8 Substitute these three values along with the original three values from steps 2 and 3 in equations 5.8 and obtain:

$$u = 1.486 \quad \sigma = 15.4^\circ$$

- 9 Compare these to the graphical solution:

$$u = 1.53 \quad \sigma = 14^\circ$$

These are a reasonable match for graphical accuracy. Vector $\mathbf{U}_1$ is link 4 of the fourbar.

- 10 Line A_1B_1 is link 3 and can be found from equation 5.2a. Line O_2O_4 is link 1 and can be found from equation 5.2b.
- 11 Check the Grashof condition, and repeat steps 4 to 7 if unsatisfied. Note that any Grashof condition is potentially acceptable in this case.
- 12 Construct a cardboard model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions.
- 13 Check transmission angles.

Input the file E05-01.4br to program FOURBAR to see Example 5-1.

5.5 SIMULTANEOUS EQUATION SOLUTION

These methods of analytical synthesis lead to sets of linear simultaneous equations. The two-position synthesis problem results in two simultaneous equations which can be solved by direct substitution. The three-position synthesis problem will lead to a system of four simultaneous linear equations and will require a more complicated method of solution. A convenient approach to the solution of sets of linear simultaneous equations is to put them in a standard matrix form and use a numerical matrix solver to obtain the answers. Matrix solvers are built into most engineering and scientific pocket calculators. Some spreadsheet packages and equation solvers will also do a matrix solution.

As an example of this general approach, consider the following set of simultaneous equations:

$$\begin{aligned} -2x_1 - x_2 + x_3 &= -1 \\ x_1 + x_2 + x_3 &= 6 \\ 3x_1 + x_2 - x_3 &= 2 \end{aligned} \quad (5.13a)$$

A system this small can be solved longhand by the elimination method, but we will put it in matrix form to show the general approach which will work regardless of the number of equations. The equations 5.13a can be written as the product of two matrices set equal to a third matrix.

$$\begin{bmatrix} -2 & -1 & 1 \\ 1 & 1 & 1 \\ 3 & 1 & -1 \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 6 \\ 2 \end{bmatrix} \quad (5.13b)$$

We will refer to these matrices as **A**, **B**, and **C**,

$$[\mathbf{A}] \times [\mathbf{B}] = [\mathbf{C}] \quad (5.13c)$$

where **A** is the matrix of coefficients of the unknowns, **B** is a column vector of the unknown terms and **C** is a column vector of the constant terms. When matrix **A** is multiplied by **B**, the result will be the same as the left sides of equation 5.13a. See any text on linear algebra such as reference 7 for a discussion of the procedure for matrix multiplication.

If equation 5.13c were a scalar equation,

$$ab = c \quad (5.14a)$$

rather than a vector (matrix) equation, it would be very easy to solve it for the unknown *b* when *a* and *c* are known. We would simply divide *c* by *a* to find *b*.

$$b = \frac{c}{a} \quad (5.14b)$$

Unfortunately, division is not defined for matrices, so another approach must be used. Note that we could also express the division in equation 5.14b as:

$$b = a^{-1}c \quad (5.14c)$$

If the equations to be solved are linearly independent, then we can find the inverse of matrix **A** and multiply it by matrix **C** to find **B**. The inverse of a matrix is defined as that matrix which when multiplied by the original matrix yields the identity matrix. The **identity matrix** is a square matrix with ones on the main diagonal and zeros everywhere else. The inverse of a matrix is denoted by adding a superscript of negative one to the symbol for the original matrix.

$$[\mathbf{A}]^{-1} \times [\mathbf{A}] = [\mathbf{I}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (5.15)$$

Not all matrices will possess an inverse. The determinant of the matrix must be nonzero for an inverse to exist. The class of problems dealt with here will yield matrices which have inverses provided that all data are correctly calculated for input to the matrix and represent a real physical system. The calculation of the terms of the inverse for a matrix is a complicated numerical process which requires a computer or preprogrammed pocket calculator to invert any matrix of significant size. A Gauss-Jordan-elimination numerical method is usually used to find an inverse. For our simple example in equation 5.13 the inverse of matrix **A** is found to be:

$$\begin{bmatrix} -2 & -1 & 1 \\ 1 & 1 & 1 \\ 3 & 1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 1.0 & 0.0 & 1.0 \\ -2.0 & 0.5 & -1.5 \\ 1.0 & 0.5 & 0.5 \end{bmatrix} \quad (5.16)$$

If the inverse of matrix **A** can be found, we can solve equation 5.13 for the unknowns **B** by multiplying both sides of the equation by the inverse of **A**. Note that unlike scalar multiplication, matrix multiplication is not commutative; i.e., **A** x **B** is not equal to **B** x **A**. We will premultiply each side of the equation by the inverse.

$$\begin{aligned} & \text{but :} & [\mathbf{A}]^{-1} \times [\mathbf{A}] \times [\mathbf{B}] &= [\mathbf{A}]^{-1} \times [\mathbf{C}] \\ & \text{so :} & [\mathbf{A}]^{-1} \times [\mathbf{A}] &= [\mathbf{I}] \\ & & [\mathbf{B}] &= [\mathbf{A}]^{-1} \times [\mathbf{C}] \end{aligned} \quad (5.17)$$

The product of **A** and its inverse on the left side of the equation is equal to the identity matrix **I**. Multiplying by the identity matrix is equivalent, in scalar terms, to multiplying by one, so it has no effect on the result. Thus the unknowns can be found by premultiplying the inverse of the coefficient matrix **A** times the matrix of constant terms **C**.

This method of solution works no matter how many equations are present as long as the inverse of **A** can be found and enough computer memory and/or time is available to do the computation. Note that it is not actually necessary to find the inverse of matrix **A** to solve the set of equations. The Gauss-Jordan algorithm which finds the inverse can also be used to directly solve for the unknowns **B** by assembling the **A** and **C** matrices into an **augmented matrix** of n rows and $n + 1$ columns. The added column is the **C** vector. This approach requires fewer calculations, so it is faster and more accurate. The augmented matrix for this example is:

$$\left[\begin{array}{ccc|c} -2 & -1 & 1 & -1 \\ 1 & 1 & 1 & 6 \\ 3 & 1 & -1 & 2 \end{array} \right] \quad (5.18a)$$

The Gauss-Jordan algorithm manipulates this augmented matrix until it is in the form shown below, in which the left, square portion has been reduced to the identity matrix and the rightmost column contains the values of the column vector of unknowns. In this case the results are $x_1 = 1$, $x_2 = 2$, and $x_3 = 3$ which are the correct solution to the original equations 5.13.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \quad (5.18b)$$

The program MATRIX, supplied with this text, solves these problems with this Gauss-Jordan elimination method and operates on the augmented matrix without actually finding the inverse of **A** in explicit form. See Appendix A for instructions on running program MATRIX. For a review of matrix algebra see reference 7.

5.6 THREE-POSITION MOTION GENERATION BY ANALYTICAL SYNTHESIS

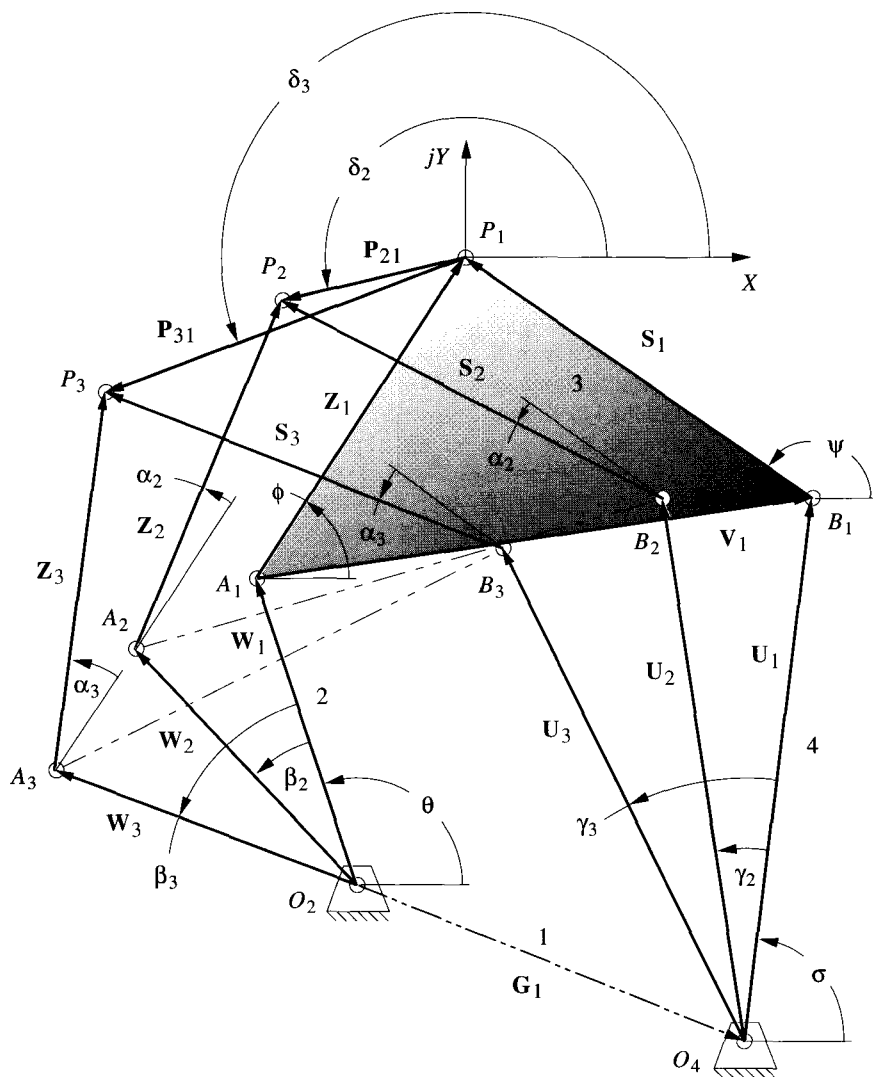
The same approach of defining two dyads, one at each end of the fourbar linkage, as used for two-position motion synthesis can be extended to three, four, and five positions in the plane. The three-position motion synthesis problem will now be addressed. Figure 5-4 shows a fourbar linkage in one general position with a coupler point located at its first precision position P_1 . Second and third precision positions (points P_2 and P_3) are also shown. These are to be achieved by the rotation of the input rocker, link 2, through as yet unspecified angles β_2 and β_3 . Note also that the angles of the coupler link 3 at each of the precision positions are defined by the angles of the position vectors $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$. The linkage shown in the figure is schematic. Its dimensions are unknown at the outset and are to be found by this synthesis technique. Thus, for example, the length of the position vector $\mathbf{Z}_1$ as shown is not indicative of the final length of that edge of link 3 nor are the lengths or angles of any of the links shown predictive of the final result.

The problem statement is:

Design a fourbar linkage which will move a line on its coupler link such that a point P on that line will be first at P_1 , later at P_2 , and still later at P_3 , and also will rotate the line through an angle α_2 between the first two precision positions and through an angle α_3 between the first and third precision positions. Find the lengths and angles of the four links and the coupler link dimensions A_1P_1 and B_1P_1 as shown in Figure 5-4.

The three-position analytical motion synthesis procedure is as follows:

For convenience, we will place the global coordinate system XY at the first precision point P_1 . We define the other two desired precision positions in the plane with respect to this global system as shown in Figure 5-4. The position difference vectors $\mathbf{P}_{21}$, drawn from P_1 to P_2 , and $\mathbf{P}_{31}$, drawn from P_1 to P_3 , have angles δ_2 and δ_3 , respectively.

**FIGURE 5-4**

Three-position analytical synthesis

The position difference vectors $\mathbf{P}_{21}$ and $\mathbf{P}_{31}$ define the displacements of the output motion of point P from point 1 to 2 and from 1 to 3, respectively.

The dyad $\mathbf{W}_1 \mathbf{Z}_1$ defines the left half of the linkage. The dyad $\mathbf{U}_1 \mathbf{S}_1$ defines the right half of the linkage. Vectors $\mathbf{Z}_1$ and $\mathbf{S}_1$ are both embedded in the rigid coupler (link 3), and both will undergo the same rotations, through angle α_2 from position 1 to position 2 and through angle α_3 from position 1 to position 3. The pin-to-pin length and angle of link 3 (vector $\mathbf{V}_1$) is defined in terms of vectors $\mathbf{Z}_1$ and $\mathbf{S}_1$ as in equation 5.2a (p. 191). The ground link is defined by equation 5.2b as before.

As we did in the two-position case, we will first solve for the left side of the linkage (vectors $\mathbf{W}_1$ and $\mathbf{Z}_1$) and later use the same procedure to solve for the right side (vectors $\mathbf{U}_1$ and $\mathbf{S}_1$). To solve for $\mathbf{W}_1$ and $\mathbf{Z}_1$ we need to now write **two vector loop equations**, one around the loop which includes positions P_1 and P_2 and the second one around the loop which includes positions P_1 and P_3 (see Figure 5-4). We will go clockwise around the first loop for motion from position 1 to 2, starting with $\mathbf{W}_2$, and then write the second loop equation for motion from position 1 to 3 starting with $\mathbf{W}_3$.

$$\begin{aligned}\mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{P}_{21} - \mathbf{Z}_1 - \mathbf{W}_1 &= 0 \\ \mathbf{W}_3 + \mathbf{Z}_3 - \mathbf{P}_{31} - \mathbf{Z}_1 - \mathbf{W}_1 &= 0\end{aligned}\quad (5.19)$$

Substituting the complex number equivalents for the vectors.

$$\begin{aligned}we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} - p_{21}e^{j\delta_2} - ze^{j\phi} - we^{j\theta} &= 0 \\ we^{j(\theta+\beta_3)} + ze^{j(\phi+\alpha_3)} - p_{31}e^{j\delta_3} - ze^{j\phi} - we^{j\theta} &= 0\end{aligned}\quad (5.20)$$

Rewriting the sums of angles in the exponents as products of terms.

$$\begin{aligned}we^{j\theta}e^{j\beta_2} + ze^{j\phi}e^{j\alpha_2} - p_{21}e^{j\delta_2} - ze^{j\phi} - we^{j\theta} &= 0 \\ we^{j\theta}e^{j\beta_3} + ze^{j\phi}e^{j\alpha_3} - p_{31}e^{j\delta_3} - ze^{j\phi} - we^{j\theta} &= 0\end{aligned}\quad (5.21a)$$

Simplifying and rearranging:

$$\begin{aligned}we^{j\theta}(e^{j\beta_2} - 1) + ze^{j\phi}(e^{j\alpha_2} - 1) &= p_{21}e^{j\delta_2} \\ we^{j\theta}(e^{j\beta_3} - 1) + ze^{j\phi}(e^{j\alpha_3} - 1) &= p_{31}e^{j\delta_3}\end{aligned}\quad (5.21b)$$

The magnitude w of vectors $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{W}_3$ is the same in all three positions because it represents the same line in a rigid link. The same can be said about vectors $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ whose common magnitude is z .

Equations 5.21 are a set of two vector equations, each of which contains two scalar equations. This set of four equations can be solved for four unknowns. The scalar equations can be revealed by substituting Euler's identity (equation 4.4a, p. 155) and separating the real and imaginary terms as was done in the two-position example above.

real part:

$$\begin{aligned}w\cos\theta(\cos\beta_2 - 1) - w\sin\theta\sin\beta_2 \\ + z\cos\phi(\cos\alpha_2 - 1) - z\sin\phi\sin\alpha_2 &= p_{21}\cos\delta_2\end{aligned}\quad (5.22a)$$

$$\begin{aligned}w\cos\theta(\cos\beta_3 - 1) - w\sin\theta\sin\beta_3 \\ + z\cos\phi(\cos\alpha_3 - 1) - z\sin\phi\sin\alpha_3 &= p_{31}\cos\delta_3\end{aligned}\quad (5.22b)$$

imaginary part (with complex operator j divided out):

$$w \sin \theta (\cos \beta_2 - 1) + w \cos \theta \sin \beta_2 + z \sin \phi (\cos \alpha_2 - 1) + z \cos \phi \sin \alpha_2 = p_{21} \sin \delta_2 \quad (5.22c)$$

$$w \sin \theta (\cos \beta_3 - 1) + w \cos \theta \sin \beta_3 + z \sin \phi (\cos \alpha_3 - 1) + z \cos \phi \sin \alpha_3 = p_{31} \sin \delta_3 \quad (5.22d)$$

There are **twelve variables** in these four equations 5.22: w , θ , β_2 , β_3 , z , ϕ , α_2 , α_3 , p_{21} , p_{31} , δ_2 , and δ_3 . **We can solve for only four.** Six of them are defined in the problem statement, namely α_2 , α_3 , p_{21} , p_{31} , δ_2 , and δ_3 . Of the remaining six, w , θ , β_2 , β_3 , z , ϕ , **we must choose two as free choices** (assumed values) in order to solve for the other four. One strategy is to assume values for the two angles, β_2 and β_3 , on the premise that we may want to specify the angular excursions of link 2 to suit some driving constraint. (This choice also has the benefit of leading to a set of linear equations for simultaneous solution.)

This leaves the magnitudes and angles of vectors **W** and **Z** to be found (w , θ , z , ϕ). To simplify the solution, we can substitute the following relationships to obtain the x and y components of the two unknown vectors **W** and **Z**, rather than their polar coordinates.

$$\begin{aligned} W_{1x} &= w \cos \theta; & Z_{1x} &= z \cos \phi \\ W_{1y} &= w \sin \theta; & Z_{1y} &= z \sin \phi \end{aligned} \quad (5.23)$$

Substituting equations 5.23 into 5.22 we obtain:

$$\begin{aligned} W_{1x} (\cos \beta_2 - 1) - W_{1y} \sin \beta_2 \\ + Z_{1x} (\cos \alpha_2 - 1) - Z_{1y} \sin \alpha_2 = p_{21} \cos \delta_2 \end{aligned} \quad (5.24a)$$

$$\begin{aligned} W_{1x} (\cos \beta_3 - 1) - W_{1y} \sin \beta_3 \\ + Z_{1x} (\cos \alpha_3 - 1) - Z_{1y} \sin \alpha_3 = p_{31} \cos \delta_3 \end{aligned} \quad (5.24b)$$

$$\begin{aligned} W_{1y} (\cos \beta_2 - 1) + W_{1x} \sin \beta_2 \\ + Z_{1y} (\cos \alpha_2 - 1) + Z_{1x} \sin \alpha_2 = p_{21} \sin \delta_2 \end{aligned} \quad (5.24c)$$

$$\begin{aligned} W_{1y} (\cos \beta_3 - 1) + W_{1x} \sin \beta_3 \\ + Z_{1y} (\cos \alpha_3 - 1) + Z_{1x} \sin \alpha_3 = p_{31} \sin \delta_3 \end{aligned} \quad (5.24d)$$

These are four equations in the four unknowns W_{1x} , W_{1y} , Z_{1x} , and Z_{1y} . By setting the coefficients which contain the assumed and specified terms equal to some constants, we can simplify the notation and obtain the following solutions.

$$\begin{aligned} A &= \cos \beta_2 - 1; & B &= \sin \beta_2; & C &= \cos \alpha_2 - 1 \\ D &= \sin \alpha_2; & E &= p_{21} \cos \delta_2; & F &= \cos \beta_3 - 1 \\ G &= \sin \beta_3; & H &= \cos \alpha_3 - 1; & K &= \sin \alpha_3 \\ L &= p_{31} \cos \delta_3; & M &= p_{21} \sin \delta_2; & N &= p_{31} \sin \delta_3 \end{aligned} \quad (5.25)$$

Substituting equations 5.25 in 5.24 to simplify:

$$AW_{1x} - BW_{1y} + CZ_{1x} - DZ_{1y} = E \quad (5.26a)$$

$$FW_{1x} - GW_{1y} + HZ_{1x} - KZ_{1y} = L \quad (5.26b)$$

$$BW_{1x} + AW_{1y} + DZ_{1x} + CZ_{1y} = M \quad (5.26c)$$

$$GW_{1x} + FW_{1y} + KZ_{1x} + HZ_{1y} = N \quad (5.26d)$$

This system can be put into standard matrix form:

$$\begin{bmatrix} A & -B & C & -D \\ F & -G & H & -K \\ B & A & D & C \\ G & F & K & H \end{bmatrix} \times \begin{bmatrix} W_{1x} \\ W_{1y} \\ Z_{1x} \\ Z_{1y} \end{bmatrix} = \begin{bmatrix} E \\ L \\ M \\ N \end{bmatrix} \quad (5.27)$$

This is of the general form of equation 5.13c. The vector of unknowns **B** can be solved for by premultiplying the inverse of the coefficient matrix **A** by the constant vector **C** or by forming the augmented matrix as in equation 5.18. For any numerical problem, the inverse of a 4 x 4 matrix can be found with many pocket calculators. The computer program MATRIX, supplied with this text, will also solve the augmented matrix equation.

Equations 5.25 and 5.26 solve the three-position synthesis problem for the left-hand side of the linkage using any pair of assumed values for β_2 and β_3 . We must repeat the above process for the right-hand side of the linkage to find vectors **U** and **S**. Figure 5-4 (p. 202) also shows the three positions of the **US** dyad, and the angles σ , γ_2 , γ_3 , ψ , α_2 , and α_3 , which define those vector rotations for all three positions. The solution derivation for the right-side dyad, **US**, is identical to that just done for the left dyad **WZ**. The angles and vector labels are the only difference. The vector loop equations are:

$$\begin{aligned} U_2 + S_2 - P_{21} - S_1 - U_1 &= 0 \\ U_3 + S_3 - P_{31} - S_1 - U_1 &= 0 \end{aligned} \quad (5.28)$$

Substituting, simplifying, and rearranging,

$$\begin{aligned} ue^{j\sigma}(e^{j\gamma_2} - 1) + se^{j\psi}(e^{j\alpha_2} - 1) &= p_{21}e^{j\delta_2} \\ ue^{j\sigma}(e^{j\gamma_3} - 1) + se^{j\psi}(e^{j\alpha_3} - 1) &= p_{31}e^{j\delta_3} \end{aligned} \quad (5.29)$$

The solution requires that two free choices be made. We will assume values for the angles γ_2 and γ_3 . Note that α_2 and α_3 are the same as for dyad **WZ**. We will, in effect, solve for angles σ and ψ by finding the x and y components of the vectors **U** and **S**. The solution is:

$$\begin{aligned}
A &= \cos \gamma_2 - 1; & B &= \sin \gamma_2; & C &= \cos \alpha_2 - 1 \\
D &= \sin \alpha_2; & E &= p_{21} \cos \delta_2; & F &= \cos \gamma_3 - 1 \\
G &= \sin \gamma_3; & H &= \cos \alpha_3 - 1; & K &= \sin \alpha_3 \\
L &= p_{31} \cos \delta_3; & M &= p_{21} \sin \delta_2; & N &= p_{31} \sin \delta_3
\end{aligned} \tag{5.30}$$

$$AU_{1x} - BU_{1y} + CS_{1x} - DS_{1y} = E \tag{5.31a}$$

$$FU_{1x} - GU_{1y} + HS_{1x} - KS_{1y} = L \tag{5.31b}$$

$$BU_{1x} + AU_{1y} + DS_{1x} + CS_{1y} = M \tag{5.31c}$$

$$GU_{1x} + FU_{1y} + KS_{1x} + HS_{1y} = N \tag{5.31d}$$

Equations 5.31 can be solved using the approach of equations 5.27 and 5.18, by changing W to U and Z to S and using the definitions of the constants given in equation 5.30 in equation 5.27.

It should be apparent that there are infinities of solutions to this three-position synthesis problem as well. An inappropriate selection of the two free choices could lead to a solution which has circuit, branch, or order problems in moving among all specified positions. Thus we must check the function of the solution synthesized by this or any other method. A simple model is the quickest check.

5.7 COMPARISON OF ANALYTICAL AND GRAPHICAL THREE-POSITION SYNTHESIS

Figure 5-5 shows the same three-position synthesis problem as was done graphically in Example 3-6 in Section 3.4. Compare this figure to Figure 3-9. The labeling has been changed to be consistent with the notation in this chapter. The points PJ , Pz , and $P3$ correspond to the three points labeled D in the earlier figure. Points $A1$, Az , and $A3$ correspond to points E ; points $B1$, Bz , and $B3$ correspond to points F . The old line AP becomes the present Z vector. Point P is the coupler point which will go through the specified precision points, PJ , Pz , and $P3$. Points A and B are the attachment points for the rockers (links 2 and 4, respectively) on the coupler (link 3). We wish to solve for the coordinates of vectors W , Z , D , and S , which define not only the lengths of those links but also the locations of the fixed pivots Oz and $O4$ in the plane and the lengths of links 3 and 1. Link 1 is defined as vector G in Figure 5-4 (p. 202) and can be found from equation 5.2b (p. 191). Link 3 is vector V found from equation 5.2a.

Four free choices must be made to constrain the problem to a particular solution out of the infinities of solutions available. In this case the values of link angles α , β , γ , and δ have been chosen to be the same values as those which were found in the graphical solution to Example 3-6 in order to obtain the same solution as a check and comparison. Recall that in doing the graphical three-position synthesis solution to this same problem we in fact also had to make four free choices. These were the x, y coordinates of the moving pivot locations E and F in Figure 3-9 which correspond in concept to our four free choices of link angles here.

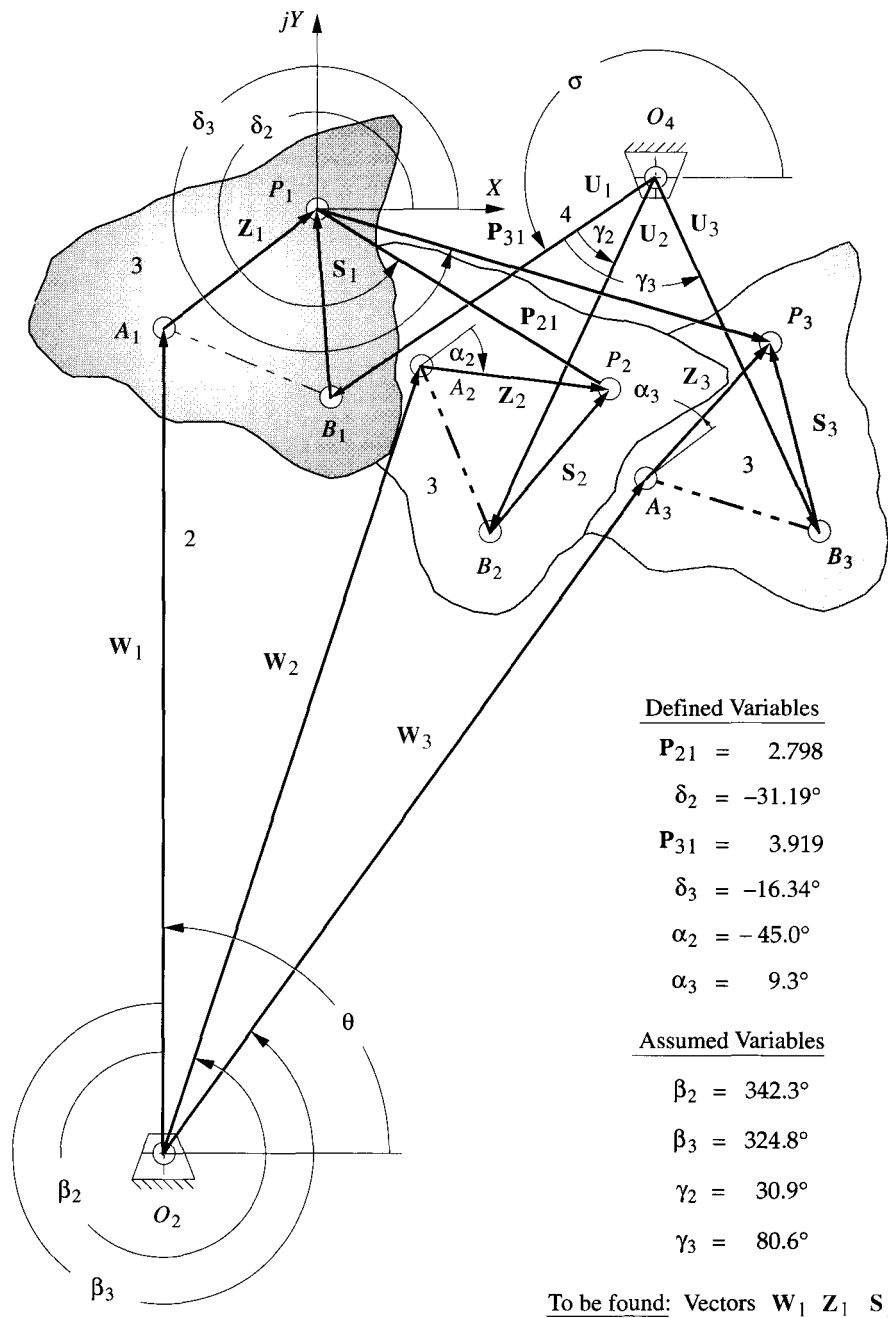


FIGURE 5-5
Data needed for three-position analytical synthesis

Example 3-5 (p. 89) also shows a graphical solution to this same problem resulting from the free choice of the x, y coordinates of points C and D on the coupler for the moving pivots (see Figure 3-8 (p. 90) and Example 3-5). We found some problems with toggle positions in that solution and redid it using points E and F as moving pivots in Example 3-6 (p. 91) and Figure 3-9 (p. 92). In effect the graphical three-position synthesis solution presented in Chapter 3 is directly analogous to the analytical solution presented here. For this analytical approach we choose to select the link angles β_2 , β_3 , γ_2 , and γ_3 rather than the moving pivot locations E and F in order to force the resulting equations to be linear in the unknowns. The graphical solution done in the earlier examples is actually a solution of nonlinear equations.

EXAMPLE 5-2

Three-Position Analytical Motion Synthesis.

Problem: Design a fourbar linkage to move the link APB shown from position $A_1P_1B_1$ to $A_2P_2B_2$ and then to position $A_3P_3B_3$.

Solution: (see Figure 5-5, p. 207)

- 1 Draw the link APB in its three desired positions, $A_1P_1B_1$, $A_2P_2B_2$, and $A_3P_3B_3$ to scale in the plane as shown in the figure.
- 2 The three positions are then defined with respect to a global origin positioned at the first precision point P_1 . The given data are the magnitudes and angles of the position difference vectors between precision points:

$$p_{21} = 2.798$$

$$\delta_{21} = -31.19^\circ$$

$$p_{31} = 3.919$$

$$\delta_{31} = -16.34^\circ$$

- 3 The angle changes of the coupler between precision points are:

$$\alpha_2 = -45^\circ$$

$$\alpha_3 = 9.3^\circ$$

- 4 The free choices assumed for the link angles are:

$$\beta_2 = 342.3^\circ$$

$$\beta_3 = 324.8^\circ$$

$$\gamma_2 = 30.9^\circ$$

$$\gamma_3 = 80.6^\circ$$

These defined variables and the free choices are also listed on the figure.

- 5 Once the free choices of link angles are made, the terms for the matrices of equation 5.27 can be defined by solving equation 5.25 for the first dyad of the linkage and equation 5.30 for the second dyad of the linkage. For this example they evaluate to:

First dyad (**WZ**):

$$A = -0.0473$$

$$B = -0.3040$$

$$C = -0.2929$$

$$D = -0.7071$$

$$E = 2.3936$$

$$F = -0.1829$$

$$G = -0.5764$$

$$H = -0.0131$$

$$K = 0.1616$$

$$L = 3.7607$$

$$M = -1.4490$$

$$N = -1.1026$$

Second dyad (US):

$$\begin{array}{llll} A = -0.1419 & B = 0.5135 & C = -0.2929 & D = -0.7071 \\ E = 2.3936 & F = -0.8367 & G = 0.9866 & H = -0.0131 \\ K = 0.1616 & L = 3.7607 & M = -1.4490 & N = -1.1026 \end{array}$$

- 6 Program MATRIX is used to solve this matrix equation once with the values from equation 5.25 inserted to get the coordinates of vectors **W** and **Z**, and a second time with values from equation 5.31 in the matrix to get the coordinates of vectors **U** and **S**. The calculated coordinates of the link vectors from equations 5.25 to 5.31 are:

$$W_x = 0.055 \quad W_y = 6.832 \quad Z_x = 1.179 \quad Z_y = 0.940$$

Link 2 = $w = 6.832$

$$U_x = -2.628 \quad U_y = -1.825 \quad S_x = -0.109 \quad S_y = 1.487$$

Link 4 = $u = 3.2$

- 7 Equation 5.2a is used to find link 3.

$$V_x = Z_x - S_x = 1.179 - (-0.109) = 1.288$$

$$V_y = Z_y - S_y = 0.940 - 1.487 = -0.547$$

Link 3 = $v = 1.399$

- 8 The ground link is found from equation 5.2b

$$G_x = W_x + V_x - U_x = 0.055 + 1.288 - (-2.628) = 3.971$$

$$G_y = W_y + V_y - U_y = 6.832 - 0.547 - (-1.825) = 8.110$$

Link 1 = $g = 9.03$

- 9 The appropriate vector components are added together to get the locations of fixed pivots O_2 and O_4 with respect to the global origin at precision point P_1 . See Figures 5-4 (p. 202) and 5-5 (p. 207).

$$O_{2x} = -Z_x - W_x = -1.179 - 0.055 = -1.234$$

$$O_{2y} = -Z_y - W_y = -0.940 - 6.832 = -7.772$$

$$O_{4x} = -S_x - U_x = -(-0.109) - (-2.628) = 2.737$$

$$O_{4y} = -S_y - U_y = -1.487 - (-1.825) = 0.338$$

Table 5-1 shows the linkage parameters as synthesized by this method. These agree with the solution found in Example 3-6 (p. 91) within its graphical accuracy. Read the diskfiles E05-02a.mat and E05-02b.mat into program MATRIX to compute these results.

This problem can also be solved with program FOURBAR using the same method as derived in Section 5.6 (p. 201). Though the derivation was done in terms of the polar

TABLE 5-1 Results of Analytical Synthesis for Example 5-2

Link Number	Analytical Solution (Length calculated, in)	Graphical Solution (Length from Fig. 3-9, p. 92, in)
1	9.03	8.9
2	6.83	6.7
3	1.40	1.5
4	3.20	3.2
Coupler Pt. =	1.51 @ 61.31 degrees	1.5 @ 61 degrees
Open/Crossed =	CROSSED	CROSSED
Start Alpha2 =	0 rad/sec ²	
Start Omega2 =	1 rad/sec	
Start Theta2 =	29 degrees	
Final Theta2 =	11 degrees	
Delta Theta2 =	-9 degrees	

coordinates of the position difference vectors $\mathbf{P}_{21}$ and $\mathbf{P}_{31}$, it was considered more convenient to supply the cartesian coordinates of these vectors to program FOURBAR. (It is generally more accurate to measure x, y coordinates from a sketch of the desired positions than to measure angles with a protractor.) Thus the program requests the rectangular coordinates of $\mathbf{P}_{21}$ and $\mathbf{P}_{31}$. For this example they are:

$$p_{21x} = 2.394 \quad p_{21y} = -1.449 \quad p_{31x} = 3.761 \quad p_{31y} = -1.103$$

The angles $\angle xz$ and $\angle x3$ must be measured from the diagram and supplied, in degrees. These six items constitute the set of "givens." **Note that these data are all relative information relating the second and third positions to the first.** No information about their absolute locations is needed. The global reference system can be taken to be anywhere in the plane. We took it to be at the first precision point position P_1 for convenience. The free choices z and 3 for the first dyad and 2 , 3 for the second dyad must also be input to program FOURBAR as they also were to program MATRIX.

Program FOURBAR then solves the matrix equation 5.27 once with the values from equation 5.25 inserted to get the coordinates of vectors $\mathbf{W}$ and $\mathbf{Z}$, and a second time with values from equation 5.31 in the matrix to get the coordinates of vectors $\mathbf{U}$ and $\mathbf{S}$. Equations 5.2 are then solved to find links 1 and 3, and the appropriate vector components are added together to get the locations of fixed pivots O_2 and O_4 . The link lengths are returned to the main part of program FOURBAR so that other linkage parameters can be calculated and the linkage animated.

Note that there are two ways to assemble any fourbar linkage, open and crossed (see Figure 4-8, p.155), and this analytical synthesis technique gives no information on which mode of assembly is necessary to get the desired solution. Thus you may have to try both modes of assembly in program FOURBAR to find the correct one after determining the proper link lengths with this method. Note also that program FOURBAR always draws

the linkage with the fixed link horizontal. Thus the animation of the solution is oriented differently than in Figure 5-5 (p. 207).

The finished linkage is the same as the one in Figure 3-8c which shows a driver dyad added to move links 2, 3, and 4 through the three precision points. You may read the diskfile E05-02.4br into program FOURBAR to see the motions of the analytically synthesized fourbar solution. The linkage will move through the three positions defined in the problem statement. The file F03-09c.6br may also be read into program SIXBAR to see the full motion of the finished sixbar linkage.

5.8 SYNTHESIS FOR A SPECIFIED FIXED PIVOT LOCATION

In Example 3-8 we used graphical synthesis techniques and inversion to create a fourbar linkage for three-position motion generation with prespecified fixed pivot locations. This is a commonly encountered problem as the available locations for fixed pivots in most machines are quite limited. Loerch et al.,^[4] show how we can use these analytical synthesis techniques to find a linkage with specified fixed pivots and three output positions for motion generation. In effect we will now take as our four free choices the x and y coordinates of the two fixed pivots instead of the angles of the links. This approach will lead to a set of nonlinear equations containing transcendental functions of the unknown angles.

Figure 5-6 shows the WZ dyad in three positions. Because we want to relate the fixed pivots of vectors W and U to our precision points, we will place the origin of our global axis system at precision point P_1 . A position vector R_1 can then be drawn from the root of vector W_1 to the global origin at P_1 , R_2 to P_2 , and R_3 to P_3 . The vector $-R_1$ defines the location of the fixed pivot in the plane with respect to the global origin at P_1 .

We will subsequently have to repeat this procedure for three positions of vector U at the right end of the linkage as we did with the three-position solution in Section 5.7. The procedure is presented here in detail only for the left end of the linkage (vectors W , Z). It is left to the reader to substitute U for W and S for Z in equations 5.32 to generate the solution for the right side.

We can write the vector loop equation for each precision position:

$$\begin{aligned} W_1 + Z_1 &= R_1 \\ W_2 + Z_2 &= R_2 \\ W_3 + Z_3 &= R_3 \end{aligned} \quad (5.32a)$$

Substitute the complex number equivalents for the vectors W_i and Z_i :

$$\begin{aligned} we^{j\theta} + ze^{j\phi} &= R_1 \\ we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} &= R_2 \\ we^{j(\theta+\beta_3)} + ze^{j(\phi+\alpha_3)} &= R_3 \end{aligned} \quad (5.32b)$$

Expand:

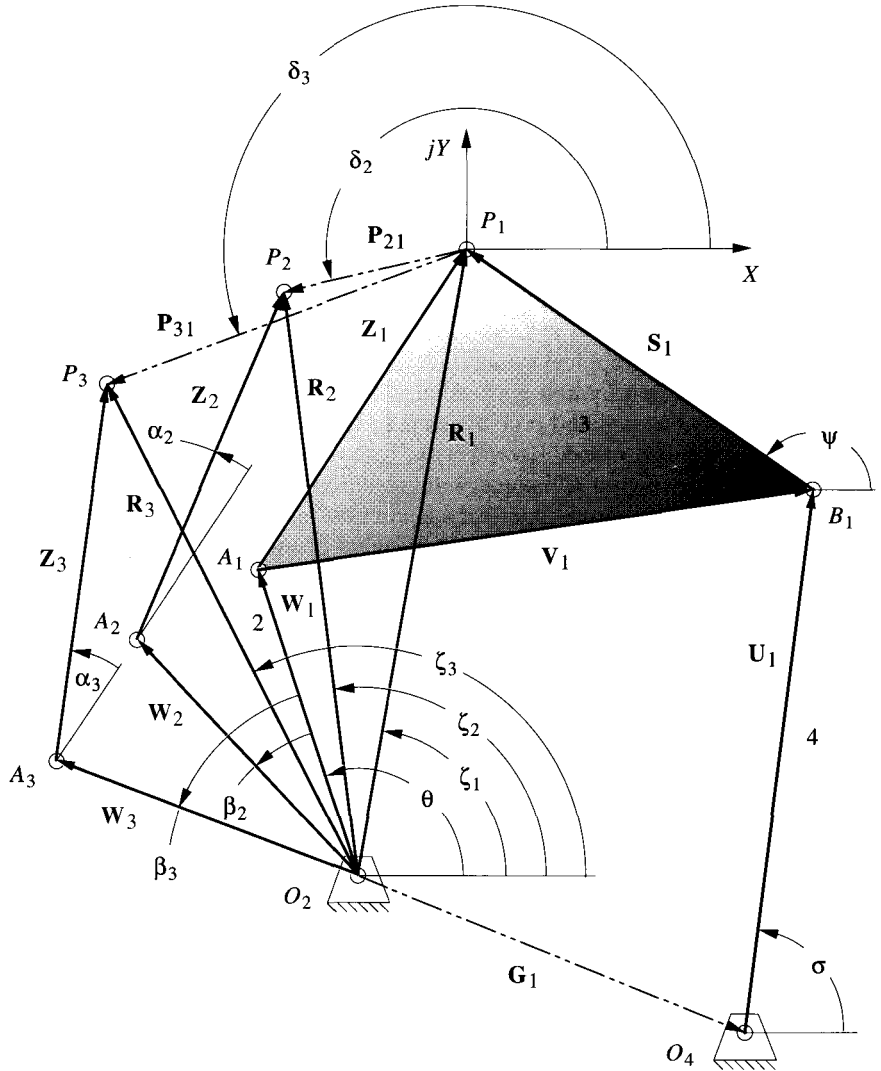


FIGURE 5-6

Three-position synthesis of a linkage with specified fixed pivot locations

$$\begin{aligned}
 we^{j\theta} + ze^{j\phi} &= \mathbf{R}_1 \\
 we^{j\theta}e^{j\beta_2} + ze^{j\phi}e^{j\alpha_2} &= \mathbf{R}_2 \\
 we^{j\theta}e^{j\beta_3} + ze^{j\phi}e^{j\alpha_3} &= \mathbf{R}_3
 \end{aligned}
 \tag{5.32c}$$

Note that:

$$\mathbf{W} = we^{j\theta}; \quad \mathbf{Z} = ze^{j\phi}
 \tag{5.32d}$$

and :

$$\begin{aligned} \mathbf{W} + \mathbf{Z} &= \mathbf{R}_1 \\ \mathbf{W}e^{j\beta_2} + \mathbf{Z}e^{j\alpha_2} &= \mathbf{R}_2 \\ \mathbf{W}e^{j\beta_3} + \mathbf{Z}e^{j\alpha_3} &= \mathbf{R}_3 \end{aligned} \quad (5.32e)$$

Previously, we chose β_2 and β_3 and solved for the vectors $\mathbf{W}$ and $\mathbf{Z}$. Now we wish to, in effect, specify the x, y components of the fixed pivot O_2 ($-R_{1x}, -R_{1y}$) as our two free choices. This leaves β_2 and β_3 to be solved for. These angles are contained in transcendental expressions in the equations. Note that, if we assumed values for β_2 and β_3 as before, there could only be a solution for $\mathbf{W}$ and $\mathbf{Z}$ if the determinant of the augmented matrix of coefficients of equations 5.32e were equal to zero.

$$\begin{bmatrix} 1 & 1 & \mathbf{R}_1 \\ e^{j\beta_2} & e^{j\alpha_2} & \mathbf{R}_2 \\ e^{j\beta_3} & e^{j\alpha_3} & \mathbf{R}_3 \end{bmatrix} = 0 \quad (5.33a)$$

Expand this determinant about the first column which contains the present unknowns β_2 and β_3 :

$$(\mathbf{R}_3e^{j\alpha_2} - \mathbf{R}_2e^{j\alpha_3}) + e^{j\beta_2}(\mathbf{R}_1e^{j\alpha_3} - \mathbf{R}_3) + e^{j\beta_3}(\mathbf{R}_2 - \mathbf{R}_1e^{j\alpha_2}) = 0 \quad (5.33b)$$

To simplify, let:

$$\begin{aligned} A &= \mathbf{R}_3e^{j\alpha_2} - \mathbf{R}_2e^{j\alpha_3} \\ B &= \mathbf{R}_1e^{j\alpha_3} - \mathbf{R}_3 \\ C &= \mathbf{R}_2 - \mathbf{R}_1e^{j\alpha_2} \end{aligned} \quad (5.33c)$$

then:

$$A + Be^{j\beta_2} + Ce^{j\beta_3} = 0 \quad (5.33d)$$

Equation 5.33d expresses the summation of vectors around a closed loop. Angles β_2 and β_3 are contained within transcendental expressions making their solution cumbersome. The procedure is similar to that used for the analysis of the fourbar linkage in Section 4.5 (p. 152). Substitute the complex number equivalents for all vectors in equation 5.33d. Expand using the Euler identity (equation 4.4a, p. 155). Separate real and imaginary terms to get two simultaneous equations in the two unknowns β_2 and β_3 . Square these expressions and add them to eliminate one unknown. Simplify the resulting mess and substitute the tangent half angle identities to get rid of the mixture of sines and cosines. It will ultimately reduce to a quadratic equation in the tangent of half the angle sought. The results are:*

$$\begin{aligned} \beta_3 &= 2 \arctan \left(\frac{K_2 \pm \sqrt{K_1^2 + K_2^2 - K_3^2}}{K_1 + K_3} \right) \\ \text{and :} \quad \beta_2 &= \arccos \left(\frac{A_5 \sin \beta_3 + A_3 \cos \beta_3 + A_6}{A_1} \right) \\ \text{or :} \quad \beta_2 &= \arcsin \left(\frac{A_3 \sin \beta_3 + A_2 \cos \beta_3 + A_4}{A_1} \right) \end{aligned} \quad (5.34a)$$

* Note also that a two-argument arctangent function must be used to obtain the proper quadrant for angle β_3 . The arccosine function returns angles in the first and second quadrants only. The arcsine function returns angles in the first and fourth quadrants only. By computing both results for β_2 one can determine which quadrant the true angle is in.

$$\begin{aligned}
 \text{where:} \quad K_1 &= A_2 A_4 + A_3 A_6 \\
 K_2 &= A_3 A_4 + A_5 A_6 \\
 K_3 &= \frac{A_1^2 - A_2^2 - A_3^2 - A_4^2 - A_6^2}{2}
 \end{aligned} \tag{5.34b}$$

$$\begin{aligned}
 \text{and:} \quad A_1 &= -C_3^2 - C_4^2; & A_2 &= C_3 C_6 - C_4 C_5 \\
 A_3 &= -C_4 C_6 - C_3 C_5; & A_4 &= C_2 C_3 + C_1 C_4 \\
 A_5 &= C_4 C_5 - C_3 C_6; & A_6 &= C_1 C_3 - C_2 C_4
 \end{aligned} \tag{5.34c}$$

$$\begin{aligned}
 C_1 &= R_3 \cos(\alpha_2 + \zeta_3) - R_2 \cos(\alpha_3 + \zeta_2) \\
 C_2 &= R_3 \sin(\alpha_2 + \zeta_3) - R_2 \sin(\alpha_3 + \zeta_2) \\
 C_3 &= R_1 \cos(\alpha_3 + \zeta_1) - R_3 \cos \zeta_3 \\
 C_4 &= -R_1 \sin(\alpha_3 + \zeta_1) + R_3 \sin \zeta_3 \\
 C_5 &= R_1 \cos(\alpha_2 + \zeta_1) - R_2 \cos \zeta_2 \\
 C_6 &= -R_1 \sin(\alpha_2 + \zeta_1) + R_2 \sin \zeta_2
 \end{aligned} \tag{5.34d}$$

The ten variables in these equations are: α_2 , α_3 , β_2 , β_3 , ζ_1 , ζ_2 , ζ_3 , R_1 , R_2 , and R_3 . The constants C_1 to C_6 are defined in terms of the eight known variables, R_1 , R_2 , R_3 , ζ_1 , ζ_2 , and ζ_3 (which are the magnitudes and angles of position vectors $\mathbf{R}_1$, $\mathbf{R}_2$, and $\mathbf{R}_3$) and the angles α_2 and α_3 which define the change in angle of the coupler. See Figure 5-6 (p. 212) for depictions of these variables.

Note in equation 5.34a that there are two solutions for each angle (just as there were to the position analysis of the fourbar linkage in Section 4.5 and Figure 4-8, p. 155). One solution in this case will be a trivial one wherein $\beta_2 = \alpha_2$ and $\beta_3 = \alpha_3$. The nontrivial solution is the one desired.

This procedure is then repeated, solving equations 5.34 for the right-hand end of the linkage using the desired location of fixed pivot O_4 to calculate the necessary angles γ_2 and γ_3 for link 4.

We have now reduced the problem to that of three-position synthesis without specified pivots as described in Section 5.6 and Example 5-2. In effect we have found the particular values of β_2 , β_3 , γ_2 , and γ_3 which correspond to the solution that uses the desired fixed pivots. The remaining task is to solve for the values of W_x , W_y , Z_x , Z_y using equations 5.25 through 5.31.



EXAMPLE 5-3

Three-Position Analytical Synthesis with Specified Fixed Pivots.

Problem: Design a fourbar linkage to move the line AP shown from position A_1P_1 to A_2P_2 and then to position A_3P_3 using fixed pivots O_2 and O_4 in the locations specified.

Solution: (see Figure 5-7)

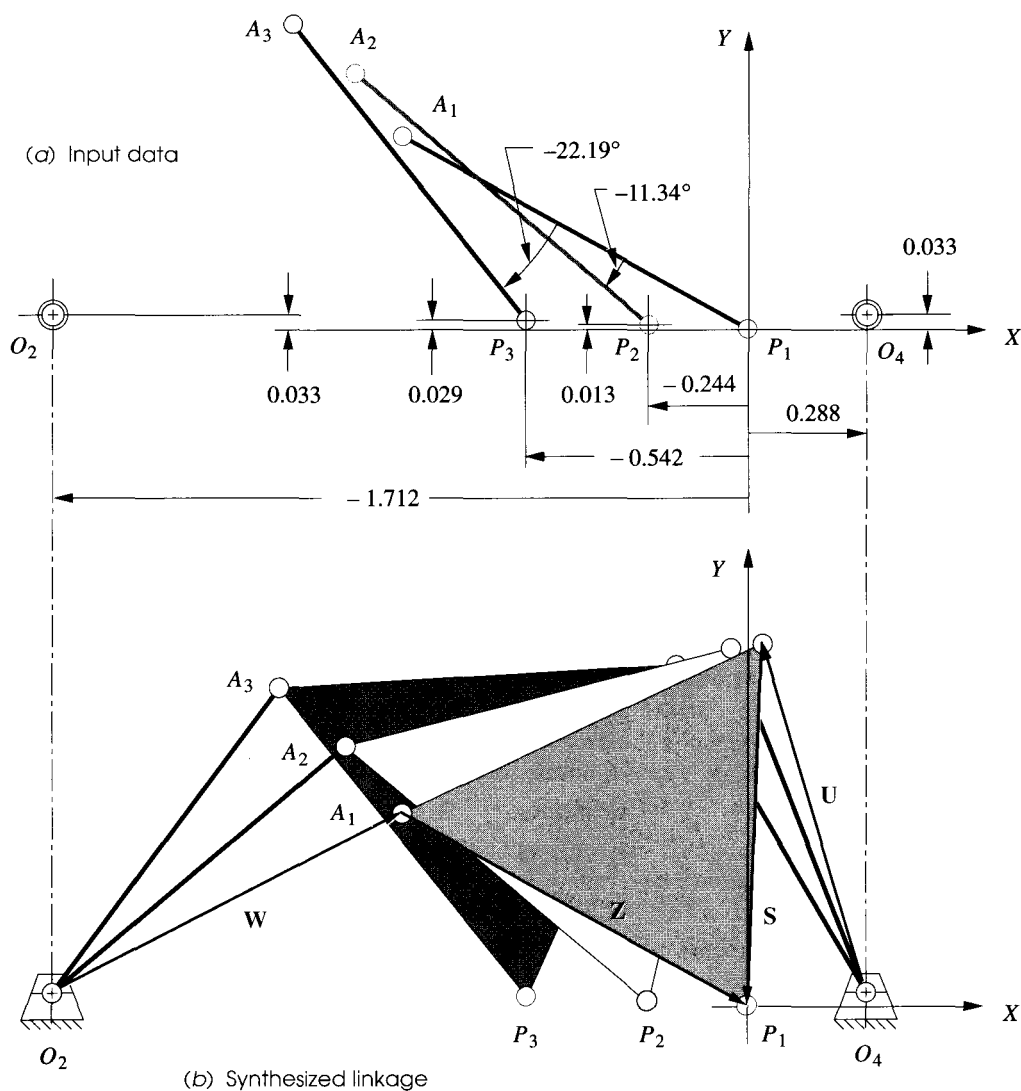


FIGURE 5-7

Three-position synthesis example for specified fixed pivots

- 1 Draw the link AP in its three desired positions, A_1P_1 , A_2P_2 , and A_3P_3 to scale in the plane as shown in Figure 5-7. The three positions are defined with respect to a global origin positioned at the first precision point P_1 . The given data are specified in parts 2 to 4 below.
- 2 The position difference vectors between precision points are:

$$P_{21x} = -0.244$$

$$P_{21y} = 0.013$$

$$P_{31x} = -0.542$$

$$P_{31y} = 0.029$$

- 3 The angle changes of the coupler between precision points are:

$$\alpha_2 = -11.34^\circ \qquad \alpha_3 = -22.19^\circ$$

- 4 The assumed free choices are the desired fixed pivot locations.

$$O_{2x} = -1.712 \qquad O_{2y} = 0.033 \qquad O_{4x} = 0.288 \qquad O_{4y} = 0.033$$

- 5 Solve equations 5.34 twice, once using the O_2 pivot location coordinates and again using the O_4 pivot location coordinates.

For pivot O_2 :

$$\begin{array}{lll} C_1 = -0.205 & C_2 = 0.3390 & C_3 = 0.4028 \\ C_4 = 0.6731 & C_5 = 0.2041 & C_6 = 0.3490 \\ A_1 = -0.6152 & A_2 = 0.0032 & A_3 = -0.3171 \\ A_4 = -0.0017 & A_5 = -0.0032 & A_6 = -0.3108 \\ K_1 = 0.0986 & K_2 = 0.0015 & K_3 = 0.0907 \end{array}$$

The values found for the link angles to match this choice of fixed pivot location O_2 are:

$$\beta_2 = 11.96^\circ \qquad \beta_3 = 23.96^\circ$$

For pivot O_4 :

$$\begin{array}{lll} C_1 = -0.3144 & C_2 = -0.0231 & C_3 = 0.5508 \\ C_4 = -0.0822 & C_5 = 0.2431 & C_6 = -0.0443 \\ A_1 = -0.3102 & A_2 = -0.0044 & A_3 = -0.1376 \\ A_4 = 0.0131 & A_5 = 0.0044 & A_6 = -0.1751 \\ K_1 = 0.0240 & K_2 = -0.0026 & K_3 = 0.0232 \end{array}$$

The values found for the link angles to match this choice of fixed pivot location O_4 are:

$$\gamma_2 = 2.78^\circ \qquad \gamma_3 = 9.96^\circ$$

- 6 At this stage, the problem has been reduced to the same one as in the previous section; i.e., find the linkage given the free choices of the above angles β_2 , β_3 , γ_2 , γ_3 , using equations 5.25 through 5.31. The data needed for the remaining calculations are those given in steps 2, 3, and 5 of this example, namely:

for dyad 1:

$$P_{21x} \quad P_{21y} \quad P_{31x} \quad P_{31y} \quad a_2 \quad a_3 \quad b_2 \quad b_3$$

for dyad 2:

$$P_{21x} \quad P_{21y} \quad P_{31x} \quad P_{31y} \quad a_2 \quad a_3 \quad g_2 \quad g_3$$

See Example 5-2 and Section 5.6 for the procedure. A matrix solving calculator, Mathcad, *TKSolver*, *Matlab*, program MATRIX, or program FOURBAR will solve this and compute the coordinates of the link vectors:

$W_x = 0.866$	$W_y = 0.500$	$Z_x = 0.846$	$Z_y = -0.533$
$U_x = -0.253$	$U_y = 0.973$	$S_x = -0.035$	$S_y = -1.006$

TABLE 5-2
Example 5-3 Results

7 The link lengths are computed as was done in Example 5-2 and are shown in Table 5-2.

This example can be read into program FOURBAR from diskfile E05-03.4br and animated.

5.9 CENTER-POINT AND CIRCLE-POINT CIRCLES

It would be quite convenient if we could find the loci of all possible solutions to the three-position synthesis problem, as we would then have an overview of the potential locations of the ends of the vectors **W**, **Z**, **U**, and **S**. Loerch^[5] shows that by holding one of the free choices (say β_2) at an arbitrary value, and then solving equations 5.25 and 5.26 while iterating the other free choice (β_3) through all possible values from 0 to 2π , a circle will be generated. This circle is the locus of all possible locations of the root of vector **W** (for the particular value of β_2 used). The root of the vector **W** is the location of the fixed pivot or center O_2 . Thus, this circle is called a **center-point circle**. The vector **N** in Figure 5-8 defines points on the *center-point* circle with respect to the global coordinate system which is placed at precision point P_1 for convenience.

If the same thing is done for vector **Z**, holding α_2 constant at some arbitrary value and iterating α_3 from 0 to 2π , another circle will be generated. This circle is the locus of all possible locations of the root of vector **Z** for the chosen value of α_2 . Because the root of vector **Z** is joined to the tip of vector **W** and **W**'s tip describes a circle about pivot O_2 in the finished linkage, this locus is called the **circle-point circle**. Vector **(-Z)** defines points on the *circle-point* circle with respect to the global coordinate system.

The *x,y* components of vectors **W** and **Z** are defined by equations 5.25 and 5.26. Negating the *x,y* components of **Z** will give the coordinates of points on the circle-point circle for any assumed value of α_2 as angle α_3 is iterated from 0 to 2π . The *x,y* components of **N** = **-Z** - **W**, define points on the O_2 center-point circle for any assumed value of β_2 as β_3 is iterated through 0 to 2π . Vector **W** is calculated using angles β_2 and β_3 , and vector **Z** using angles α_2 and α_3 , both from equations 5.25 and 5.26.

For the right-hand dyad, there will also be separate center-point circles and circle-point circles. The *x,y* components of **M** = **-S** - **U**, define points on the O_4 center-point circle for any assumed value of γ_2 as γ_3 is iterated through 0 to 2π . (See Figure 5-8 and also Figure 5-4, p. 202.) Negating the *x,y* components of **S** will give the coordinates of points on the circle-point circle for any assumed value of ψ_2 as ψ_3 is iterated through 0 to 2π . Vector **U** is calculated using angles γ_2 and γ_3 , and vector **S** using angles ψ_2 and ψ_3 , both from equations 5.30 and 5.31.

Note that there is still an infinity of solutions because we are choosing the value of one angle arbitrarily. Thus there will be an **infinite number of sets of center-point and circle-point circles**. A computer program can be of help in choosing a linkage design which has pivots in convenient locations. Program FOURBAR, provided with this text, will calculate the solutions to the analytical synthesis equations derived in this section, for user-selected values of all the free choices needed for three-position synthesis, both

Link 1 = 2.00 in
Link 2 = 1.00 in
Link 3 = 1.00 in
Link 4 = 1.01 in
Coupler Pt. = 1.0 in @ -60.73°
Circuit = Open
Start Theta2 = 30°
Final Theta2 = 54°
Delta Theta2 = 12°

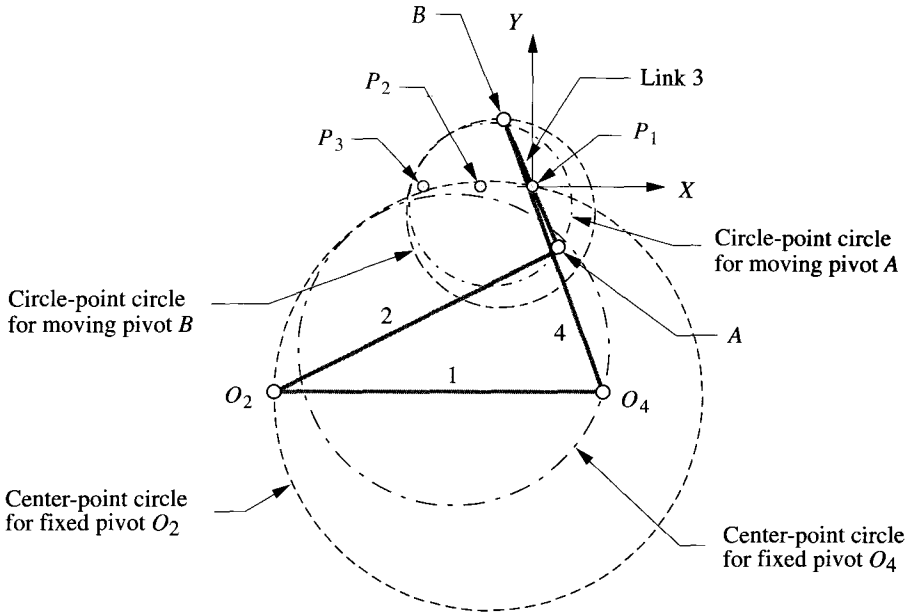


FIGURE 5-9

Circle-point and center-point circles and a linkage that reaches the precision points

tions I_{23} and I_{34} . Note that the coordinate system has its origin at the reference precision point, in this case P_1 , from which we measured all parameters used in the analysis. These circles define the pivot loci of all possible linkages which will reach the three precision points P_1 , P_2 , and P_3 that were specified for particular choices of angles β_2 , α_2 , γ_2 , and ψ_2 . An example linkage is drawn on the diagram to illustrate one possible solution.

5.10 FOUR- AND FIVE-POSITION ANALYTICAL SYNTHESIS

The same techniques derived above for two- and three-position synthesis can be extended to four and five positions by writing more vector loop equations, one for each precision point. To facilitate this we will now put the vector loop equations in a more general form, applicable to any number of precision positions. Figure 5-4 (p. 202) will still serve to illustrate the notation for the general solution. The angles α_2 , α_3 , β_2 , β_3 , γ_2 , and γ_3 will now be designated as α_k , β_k , and γ_k , $k = 2$ to n , where k represents the precision position and $n = 2, 3, 4$, or 5 represents the total number of positions to be solved for. The vector loop general equation set then becomes:

$$\mathbf{W}_k + \mathbf{Z}_k - \mathbf{P}_{k1} - \mathbf{Z}_1 - \mathbf{W}_1 = 0, \quad k = 2 \text{ to } n \quad (5.35a)$$

Which, after substituting the complex number forms and simplifying becomes:

$$we^{j\theta}(e^{j\beta_k} - 1) + ze^{j\phi}(e^{j\alpha_k} - 1) = p_{k1}e^{j\delta_k}, \quad k = 2 \text{ to } n \quad (5.35b)$$

This can be put in a more compact form by substituting vector notation for those terms to which it applies, let:

$$\mathbf{W} = we^{j\theta}; \quad \mathbf{Z} = ze^{j\phi}; \quad \mathbf{P}_{k1} = p_{k1}e^{j\delta_k} \quad (5.35c)$$

then:

$$\mathbf{W}(e^{j\beta_k} - 1) + \mathbf{Z}(e^{j\alpha_k} - 1) = \mathbf{P}_{k1}e^{j\delta_k}, \quad k = 2 \text{ to } n \quad (5.35d)$$

Equation 5.35d is called the *standard form equation* by Erdman and Sandor.^[6] By substituting the values of α_k , β_k , and δ_k , in equation 5.35d for all the precision positions desired, the requisite set of simultaneous equations can be written for the left dyad of the linkage. The standard form equation applies to the right-hand dyad **US** as well, with appropriate changes to variable names as required.

$$\mathbf{U}(e^{j\beta_k} - 1) + \mathbf{S}(e^{j\alpha_k} - 1) = \mathbf{P}_{k1}e^{j\delta_k}, \quad k = 2 \text{ to } n \quad (5.35e)$$

The number of resulting equations, variables, and free choices for each value of n is shown in Table 5-3 (after Erdman and Sandor). They provide solutions for the four- and five-position problems in reference 6. The circle-point and center-point circles of the three-position problem become cubic curves, called **Burmester curves**, in the four-position problem. Erdman's commercially available computer program LINCAGES^[8] solves the **four-position problem** in an interactive way, allowing the user to select center and circle pivot locations on their Burmester curve loci, which are drawn on the graphics screen of the computer.

5.11 ANALYTICAL SYNTHESIS OF A PATH GENERATOR WITH PRESCRIBED TIMING

The approach derived above for motion generation synthesis is also applicable to the case of **path generation with prescribed timing**. In path generation, the precision points are to be reached, but the angle of a line on the coupler is not of concern. Instead, the timing at which the coupler reaches the precision point is specified in terms of input rocker angle β_2 . In the three-position motion generation problem we specified the angles α_2 and α_3 of vector $\mathbf{Z}$ in order to control the angle of the coupler. Here we instead want to specify angles β_2 and β_3 of the input rocker, to define the timing. Before, the free choices were β_2 and β_3 . Now they will be α_2 and α_3 . In either case, all four angles are either specified or assumed as free choices and the solution is identical. Figure 5-4 (p. 202) and equations 5.25, 5.26, 5.30, and 5.31 apply to this case as well. This case can be extended to as many as five precision points as shown in Table 5-3.

5.12 ANALYTICAL SYNTHESIS OF A FOURBAR FUNCTION GENERATOR

A similar process to that used for the synthesis of path generation with prescribed timing can be applied to the problem of function generation. In this case we do not care about motion of the coupler at all. In a fourbar function generator, the coupler exists only to **couple** the input link to the output link. Figure 5-10 shows a fourbar linkage in three

TABLE 5-3 Number of Variables and Free Choices for Analytical Precision Point Motion and Timed Path Synthesis.⁷

No. of Positions (n)	No. of Scalar Variables	No. of Scalar Equations	No. of Prescribed Variables	No. of Free Choices	No. of Available Solutions
2	8	2	3	3	∞^3
3	12	4	6	2	∞^2
4	16	6	9	1	∞^1
5	20	8	12	0	Finite

positions. Note that the coupler, link 3, is merely a line from point *A* to point *P*. Point *P* can be thought of as a coupler point which happens to coincide with the pin joint between links 3 and 4. As such it will have simple arc motion, pivoting about *O*₄, rather than, for example, the higher-order path motion of the coupler point *P*₁ in Figure 5-4.

Our **function generator** uses link 2 as the input link and takes the output from link 4. The “function” generated is the **relationship between the angles of link 2 and link 4** for the specified three-position positions, *P*₁, *P*₂, and *P*₃. These are located in the plane with respect to an arbitrary global coordinate system by position vectors **R**₁, **R**₂, and **R**₃. The function is:

$$\gamma_k = f(\beta_k), \qquad k = 1, 2, \dots, n; \qquad n \leq 7 \qquad (5.36)$$

This is not a continuous function. The relationship holds only for the discrete points (*k*) specified.

To synthesize the lengths of the links needed to satisfy equation 5.36, we will write vector loop equations around the linkage in pairs of positions, as was done for the previous examples. However, we now wish to include both link 2 and link 4 in the loop, since link 4 is the output. See Figure 5-10.

$$\begin{aligned} \mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{U}_2 + \mathbf{U}_1 - \mathbf{Z}_1 - \mathbf{W}_1 &= 0 \\ \mathbf{W}_3 + \mathbf{Z}_3 - \mathbf{U}_3 + \mathbf{U}_1 - \mathbf{Z}_1 - \mathbf{W}_1 &= 0 \end{aligned} \qquad (5.37a)$$

rearranging:

$$\begin{aligned} \mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{Z}_1 - \mathbf{W}_1 &= \mathbf{U}_2 - \mathbf{U}_1 \\ \mathbf{W}_3 + \mathbf{Z}_3 - \mathbf{Z}_1 - \mathbf{W}_1 &= \mathbf{U}_3 - \mathbf{U}_1 \end{aligned} \qquad (5.37b)$$

but,

$$\begin{aligned} \mathbf{P}_{21} &= \mathbf{U}_2 - \mathbf{U}_1 \\ \mathbf{P}_{31} &= \mathbf{U}_3 - \mathbf{U}_1 \end{aligned} \qquad (5.37c)$$

substituting:

$$\begin{aligned} \mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{Z}_1 - \mathbf{W}_1 &= \mathbf{P}_{21} \\ \mathbf{W}_3 + \mathbf{Z}_3 - \mathbf{Z}_1 - \mathbf{W}_1 &= \mathbf{P}_{31} \end{aligned} \qquad (5.37d)$$

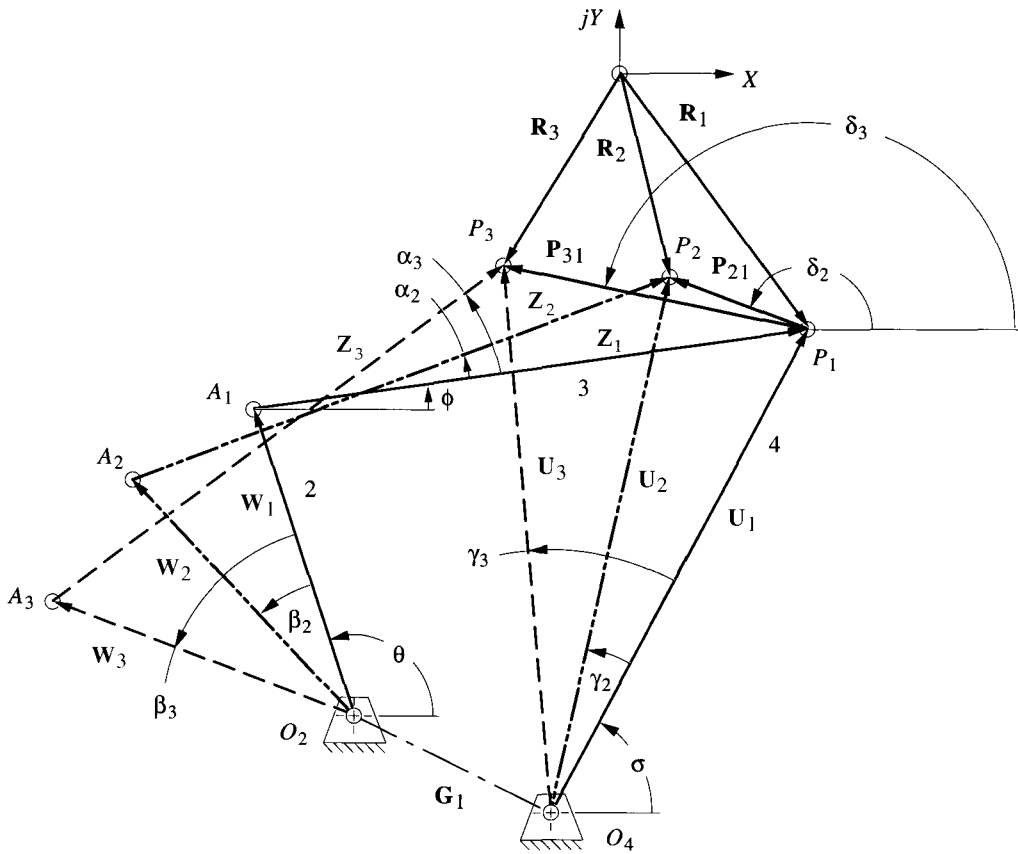


FIGURE 5-10

Analytical synthesis of a fourbar function generator

$$\begin{aligned}
 we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} - ze^{j\phi} - we^{j\theta} &= p_{21}e^{j\delta_2} \\
 we^{j(\theta+\beta_3)} + ze^{j(\phi+\alpha_3)} - ze^{j\phi} - we^{j\theta} &= p_{31}e^{j\delta_3}
 \end{aligned}
 \quad (5.37e)$$

Note that equations 5.37d and 5.37e are identical to equations 5.19 and 5.20 derived for the three-position motion generation case and can also be put into Erdman's *standard form*^[6] of equation 5.35 for the n -position case. The twelve variables in equation 5.37e are the same as those in equation 5.20: w , θ , β_2 , β_3 , z , ϕ , α_2 , α_3 , p_{21} , p_{31} , δ_2 , and δ_3 .

For the three-position function generation case the solution procedure then can be the same as that described by equations 5.20 through 5.27 for the motion synthesis problem. In other words, the solution equations are the same for **all three types** of kinematic synthesis, *function generation*, *motion generation*, and *path generation with prescribed timing*. This is why Erdman and Sandor called equation 5.35 the *standard form equation*. To develop the data for the function generation solution, expand equation 5.37b:

$$\begin{aligned} w e^{j(\theta+\beta_2)} + z e^{j(\phi+\alpha_2)} - z e^{j\phi} - w e^{j\theta} &= u e^{j(\sigma+\gamma_2)} - u e^{j\sigma} \\ w e^{j(\theta+\beta_3)} + z e^{j(\phi+\alpha_3)} - z e^{j\phi} - w e^{j\theta} &= u e^{j(\sigma+\gamma_3)} - u e^{j\sigma} \end{aligned} \quad (5.37f)$$

There are also **twelve variables** in equation 5.37f: $w, \theta, z, \phi, \alpha_2, \alpha_3, \beta_2, \beta_3, u, \sigma, \gamma_2$, and γ_3 . We can solve for any four. Four angles, $\beta_2, \beta_3, \gamma_2$, and γ_3 are specified from the function to be generated in equation 5.36. This leaves **four free choices**. In the function generation problem it is often convenient to define the length of the output rocker, u , and its initial angle σ to suit the package constraints. Thus, selecting the components u and σ of vector U_1 can provide two convenient free choices of the four required.

With u, σ, γ_2 , and γ_3 known, U_2 and U_3 can be found. Vectors P_{21} and P_{31} can then be found from equation 5.37c. Six of the unknowns in equation 5.37e are then defined, namely, $\beta_2, \beta_3, p_{21}, p_{31}, \delta_2$, and δ_3 . Of the remaining six ($w, \theta, z, \phi, \alpha_2, \alpha_3$), we must assume values for two more as free choices in order to solve for the remaining four. We will assume values (free choices) for the two angles, α_2 and α_3 (as was done for path generation with prescribed timing) and solve equations 5.37e for the components of W and Z (w, θ, z, ϕ). We have now reduced the problem to that of Section 5.6 and Example 5-2. See equations 5.20 through 5.27 for the solution.

Having chosen vector U_1 (u, σ) as a free choice in this case, we only have to solve for one dyad, WZ . Though we arbitrarily choose the length of vector U_1 , the resulting function generator linkage can be scaled up or down to suit packaging constraints without affecting the input/output relation defined in equation 5.36, because it is a function of angles only. This fact is not true for the motion or path generation cases, as scaling them will change the absolute coordinates of the path or motion output precision points which were specified in the problem statement.

Table 5-4 shows the relationships between number of positions, variables, free choices, and solutions for the function generation case. Note that up to seven angular output positions can be solved for with this method.

TABLE 5-4 Number of Variables and Free Choices for Function Generation Synthesis.⁷

No. of Positions (n)	No. of Scalar Variables	No. of Scalar Equations	No. of Prescribed Variables	No. of Free Choices	No. of Available Solutions
2	8	2	1	5	∞^5
3	12	4	4	4	∞^4
4	16	6	7	3	∞^3
5	20	8	10	2	∞^2
6	24	10	13	1	∞^1
7	28	12	16	0	Finite